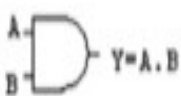
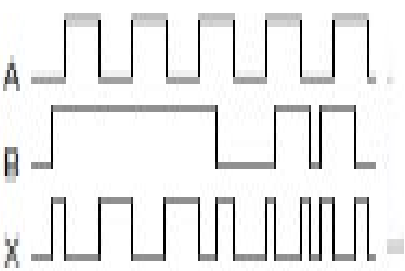
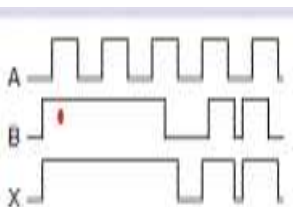
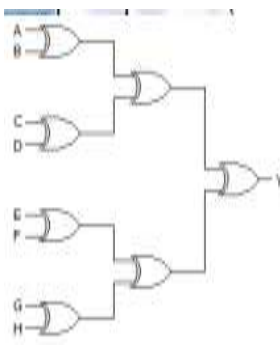
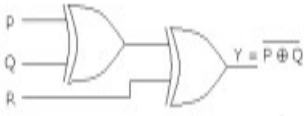
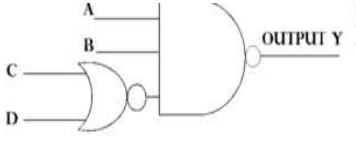
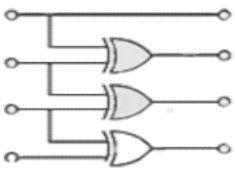
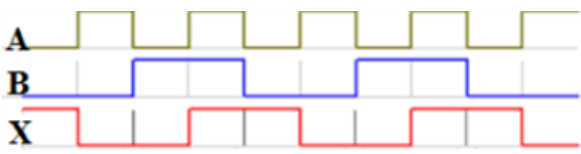


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Digital Electronics (DE) Practice Book (Sem-III 2026)								
Note: This Practice Book is only for reference purpose. LJU Test question paper may not be completely set from Practice Book.								
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
1	1	The given hexadecimal number (1E.53)16 is equivalent to _____.	B	1	(35.684)8	(36.246)8	(34.340)8	(35.599)8
2	1	The octal number (651.124)8 is equivalent to _____.	A	1	(1A9.2A)16	(1B0.10)16	(1A8.A3)16	(1B0.B0)16
3	1	The octal equivalent of the decimal number (417)10 is ____.	A	1	(641)8	(619)8	(640)8	(598)8
4	1	(170)10 is equivalent to _____.	C	1	(FD)16	(DF)16	(AA)16	(AF)16
5	1	Convert (214)8 into decimal.	A	1	(140)10	(141)10	(142)10	(130)10
6	1	Convert (0.345)10 into an octal number.	B	1	(0.16050)8	(0.26050)8	(0.19450)8	(0.24040)8
7	1	Convert the binary number (01011.1011)2 into decimal.	A	1	(11.6875)10	(11.5874)10	(10.9876)10	(10.7893)10
8	1	Convert binary to octal: (110110001010)2 =?	B	1	(5512)8	(6612)8	(4532)8	(6745)8
9	1	Which of the following is not a positional number system?	A	1	Roman Number System	Octal Number System	Binary Number System	Hexadecimal Number System
10	1	The value of radix in binary number system is _____.	D	1	4	1	10	2
11	1	The binary equivalent of the decimal number 10 is _____.	C	1	10	100	1010	101
12	1	The octal equivalent of 1100101.001010 is ____.	B	1	624.12	145.12	154.12	145.21
13	1	The hexadecimal number for (95.5)10 is ____.	(5F.8) 16	1				
14	1	The hexadecimal number ‘A0’ has the decimal value equivalent to ____.	160	1				
15	1	The decimal equivalent of hex number 1A53 is ____.	6739	1				
16	1	_____ be the decimal equivalent of 111011.10.	59.5	1				
17	1	Given that (16)10 = (100)x, find the value of x.	D	2	X=10	X=1	X=8	X=4
18	1	(4433)5 = ()10 = ()2		2				
19	1	1) (673.124)8 = ()2 2) (4522.25)10 = ()2 3) (FACE)16 = ()10 4) (10101010)2 = ()8 = ()16		4				
20	1	Convert following Octal Number to Hexadecimal and Binary 414, 574, 725.25.	(10C,17C,1D5.54)16 , (100001100, 101111100, 111010101.010101)2	3				
21	1	Convert the number (435)7into equivalent radix-10 and radix-4 number system.		3				
22	1	Do as directed: (10110100)2= (?)gray = (?)XS-3 = (?)BCD		3				
23	1	(i) Convert (75)10 = (_____)2 (ii) Convert (101011)2 = (_____)10 (iii)Convert (10101101)2 = (_____)16 = (_____)8		3				
24	1	Convert the decimal number 225.225 to binary, octal and hexadecimal.		3				
25	1	Convert following numbers. (a)(4021.2)5 = ()10 (b) (B65F)16= ()10 (c) (630.4)8 = ()10 (d) (41)10 = ()8		2				
26	1	Convert the following Hexadecimal numbers to Octal. (a) 4F7.A8 (b) BC70.0E (c) 42FD		3				
27	1	Convert following Hexadecimal Number to Decimal B28, FFF, F28	(2856,4095,3880)10	2				
28	1	(1011011101101110)2 = ()16	(B76E)16	1				
29	1	Convert the decimal number 187 to 8-bit binary.	A	1	(10111011)2	(11011101)2	(10111101)2	(10111100)2
30	1	The base (or radix) of the number system such that the following equation holds is (312/20) = 13.1	A	1	5	4	3	2
31	1	Convert (BABA)16into equivalent base 5,base 6, base 7 Number System.		3				
32	1	Convert the decimal number 250.5 to base 3, base 4, base 7 and base 8 .		4				
33	1	Convert (4BAC)16= ()8 = ()4 = ()2 = ()10. Show all steps of conversion.		2				
34	1	Express decimal number 60.875 into binary form.		1				
35	1	Convert the following numbers form given base to the base indicates. 1. (AEF2.B6)16 = (_____)2 2. (674.12)8 = (_____)10 3. (110110.1011)2 = (_____)16		2				
36	1	Express decimal number 10.875 into binary form.		2				
37	1	Convert the Decimal Number 412.5 to base 3, 4 and 7		3				
38	1	Nibble means_____	B	1	2 bit	4 bit	8 bit	16 bit
39	1	Convert the binary number (1001.0010)2 to decimal.	B	2	90.125	9.125	125	12.5
40	1	i. Convert (FFFF)16=(_____)10. ii. Convert (125.625)10=(_____)2.		2				
41	1	(52)10 = ()2	(110100)2	1				
42	1	(436)8 = ()16		1				
43	1	(5C7)16 = ()10	(1479)10	1				
44	1	(11011.101)2 = ()10	(27.625)10	1				
45	1	Convert following 1. (4E7.2)16 = (?)8 2. (521.3)8 = (?)2		2				
46	1	Convert (10101101)2 = (_____)16 = (_____)8		2				
47	1	(734)8 = ()16	D	1	C1D	DC1	1CD	1DC
48	1	(1111.11)2 = (?)8 = (?)10		2				
49	1	(AEF2.B6)16 = (_____)2		1				
50	1	(674.12)8 = (_____)10		1				
51	1	(110110.1011)2 = (_____)16		1				
52	1	(101001011)2 = (_____)10		1				
53	1	(11101110)2 = (_____)8		1				
54	1	(68)10 = (_____)16		1				
55	1	Consider the equation (123)5 = (x8)y with x and y as unknown. The number of possible solutions is _____.	3	1				
56	1	(1217)8 is equivalent to _____	C	1	(1217)16	(2297)10	(028F)16	(0B17)16
57	1	(73)x (in base x number system) is equal to (54)y (in base y number system), the possible values of x and y are____ , _____	D	2	8,16	8,10	10,12	8,11
58	1	Which of the following number is not allowed in radix – 7 (base 7) system?	B	1	666	739	461	124
59	1	If (2.3)4 + (1.2)4 = y4, then value of y in base 4 system.	A	1	10.1	10.01	10.2	1.02

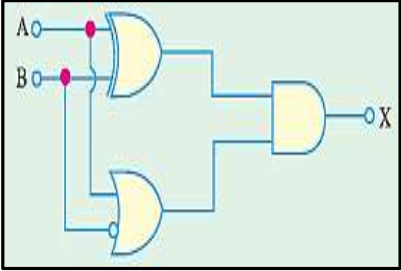
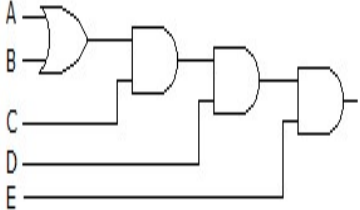
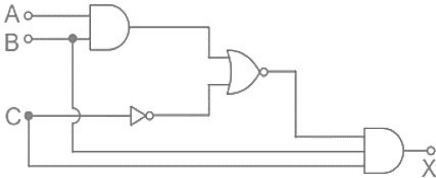
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Digital Electronics (DE) Practice Book (Sem-III 2026)								
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Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
60	1	If $(2.3)_4 + (1.2)_4 = Y_{10}$, then value of Y in base 10 system?	A	2	4.25	10.1	10.01	3.5
61	1	The Octal equivalent of hexadecimal number AB.CD is _____	A	1	253.632	253.314	526.314	526.632
62	1	Which number system has a base 16?	D	1	Octal	Binary	Decimal	Hexadecimal
63	1	What is Digital Electronics?	D	1	Engineering of devices that produce digital signal	Field of electronics involving the study of digital signal	Engineering of devices that accepts digital signals	All of the mentioned
64	1	Convert the following numbers as directed: $(130)_{10} = ()_2$	B	1	$(10000011)_2$	$(10000010)_2$	$(10000111)_2$	$(111001)_2$
65	1	Convert the following numbers as directed: $(1011011)_2 = ()_{10}$	D	1	$(91)_8$	$(52)_{10}$	$(41)_{10}$	$(91)_{10}$
66	1	Convert $(B65F)_{16} = ()_{10}$.	C	1	$(44526)_{10}$	$(78864)_{10}$	$(46687)_{10}$	$(48756)_{10}$
67	1	Convert $(41)_{10} = ()_2$.	D	1	$(110011)_2$	$(111000)_2$	$(101010)_2$	$(101001)_2$
68	1	Convert decimal number $(43)_{10}$ to binary.	C	1	$(110110)_2$	$(101010)_2$	$(101011)_2$	$(111001)_2$
69	1	Convert octal number $(234)_8$ to hexadecimal.	A	1	$(9C)_{16}$	$(8C)_{16}$	$(9C)_8$	$(9C)_2$
70	1	Convert decimal number $(0.252)_{10}$ to binary.	B	1	$(0.00001)_2$	$(0.010000)_2$	$(1.010000)_2$	$(10.00100)_2$
71	1	Convert the Hexadecimal numbers to Octal: 4F7.A8		2				
72	1	Convert the Hexadecimal numbers to Octal: BC70.0E		2				
73	1	A number $(1217)_8$ can be represented as _____	D	1	$(656)_{10}$	$(001010001111)_2$	$(28F)_{16}$	Both B and C
74	1	The number $(100000)_2$ would appear just immediately after –	D	1	$(37)_8$	$(1F)_{16}$	$(11111)_2$	All A,B and C
75	1	$(F23.23)_{16} = (?)_8 = (?)_2$	C	1	7442.106, 111100100011.001000 11	7442.106, 111110100011.0010 0011	7443.106, 111100100011.00100 011	7443.106, 111100000001.00100 011
76	1	If $(154)_b / (14)_b = (8)_{10}$ then what is the value of radix ‘b’ ?	B	1	10	7	12	8
77	1	How many number of ‘1’ would be appeared in binary representation of - $3 \times 512 + 5 \times 64 + 7 \times 8 + 3$?	D	1	7	5	6	9
78	1	Convert (DEAD) ₁₆ into equivalent radix-10 and radix-8 number system.		2				
79	1	Consider the equation $(129)_5 = (x9)_y$ with x and y as unknown. The number of possible solutions is _____.	A	1	not possible	5	3	4
80	1	convert (FFFF) ₁₆ to decimal & octal numbers respectively.	D	1	65235 & 177787	62325 & 177878	65335 & 177878	65535 & 177777
81	1	The binary equivalent of $(11.6275)_{10}$ is equals to _____.	D	1	101.11011	1011.1001	101.0011	1011.101
82	1	Convert the hexadecimal number (9999.9999) to octal number.	D	1	1414631.463	463144.1146	141361.1146	114631.4631
83	1	Which of the following is equivalent to decimal number of (BFA0) ₁₆ hexadecimal number ?	C	1	90456	40956	49056	56049
84	1	Consider the following statements i) octal number is 1/4th the length of corresponding binary number ii) Hexadecimal is 1/3rd length of corresponding binary number iii) In 10 bit binary number, 512 is the weight of 2nd digit from MSB. Which of statement/s is/are not correct from above ?	A	1	i,ii,iii	i, ii only	i, iii only	ii, iii only
85	1	Convert the following numbers as per required: (i) $(965.198)_{10} = (?)_{16}$ (ii) $(3B.4)_{16} = (?)_8$ (iii) $(1110011100.110001)_2 = (?)_{10} = (?)_{16}$		3				
86	1	Convert the decimal number 279.5 to base 3 and 7.		2				
87	1	The hexadecimal equivalent of largest binary number with 14-bits is?	A	1	3FFF	7FFF	8FFF	None of Above
88	2	The 2’s complement of the number 1101101 is _____.	10011	1				
89	2	2's complement of any binary number can be calculated by adding 1's complement twice.	False	1				
90	2	The 2’s complement of the number 1101110 is _____.	0010010	1				
91	2	Subtract $(45)_8$ from $(66)_8$ using 2's complement method.		2				
92	2	Subtract $(45)_{10}$ from $(93)_{10}$ using 1’s Complement Method.	$(48)_{10}$	2				
93	2	Subtract using 2’s complement method. $(10010 - 10011)_2$		2				
94	2	Perform subtraction of $(78 - 58)_{10}$ using 2’s complement method.		2				
95	2	Using 10’s complement, subtract : $(72532-3250)_{10}$		2				
96	2	Using 2’s complement, subtract : $(1010100 - 1000100)_2$		2				
97	2	Perform the subtraction with the following numbers using 1’s complement and 2’s complements: (a) 11010-1101, (b) 10010-10011		4				
98	2	Perform the operation of subtractions with the following binary number using 2’s complement (i) 10010 – 10011 (ii) 100 – 110000 (iii) 11010 – 10000		6				
99	2	Find the 10’s complement of the following: (1) $(6106)_{10}$ (2) $(935)_{10}$		4				
100	2	Perform following subtraction using 2’s complement method. $(11010)_2 - (10000)_2$		2				
101	2	Subtract $(32)_{10}$ from $(58)_{10}$ using 8-bit 2's complement arithmetic.		2				
102	2	Subtract 14 from 46 using 8-bit 2's complement arithmetic.		2				
103	2	Subtract 27.50 from 68.75 using 12-bit 1's complement arithmetic.		2				
104	2	Add the following binary numbers: 1. 11011 + 1101 2. 1010.11 + 1101.0 + 1001.11+1111.11		2				
105	2	Subtract the following binary numbers. 1. 1100.10-111.01 2. 10110-1011		2				
106	2	The addition of these binary numbers 101001+ 010011 would generate:	C	1	101110	000111	111100	010100
107	2	The 1's complement of a binary number is obtained by changing ____	C	1	Each 1 to a 0	Each 0 to 1	Each 1 to 0 and each 0 to 1	None of Above

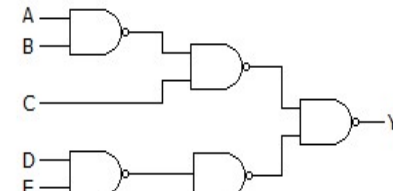
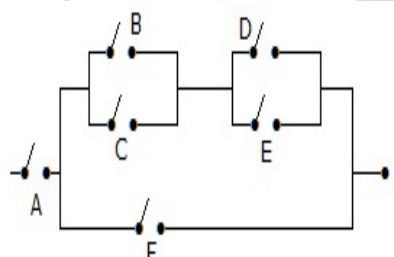
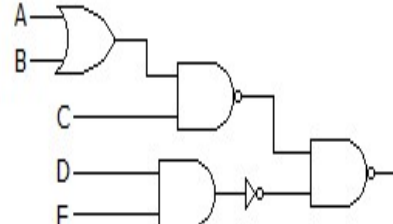
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
108	2	Subtract the following hexadecimal numbers using the 1's complement arithmetic. (i) $48_{16} - 26_{16}$ (ii) $45_{16} - 74_{16}$		4				
109	2	Subtract the following hexadecimal numbers using the 2's complement arithmetic. (i) $69_{16} - 43_{16}$ (ii) $27_{16} - 73_{16}$		4				
110	2	Subtract the following decimal numbers using the 9's and 10's complement arithmetic. (i) $274_{10} - 86_{10}$ (ii) $376.3_{10} - 765.6_{10}$		4				
111	2	Subtract the following decimal numbers using 2's complement arithmetic: 1. $46 - 19$ 2. $36.75 - 89.5$		4				
112	2	Subtract the following numbers using 9's complement: $745.81 - 436.62$		2				
113	2	Subtract the following numbers using 10's complement: $416.73 - 2928.54$		2				
114	2	Express -45 and -73.75 in 2's complement form.		2				
115	2	Add 47.25 to 55.75 using binary arithmetic.		2				
116	2	Add -75 to +26 using 8 bit 2's complement arithmetic.		2				
117	2	Subtract the following decimal numbers using 1's complement arithmetic: 1. $46-84$ 2. $63.75-17.5$		4				
118	2	Find out Y, if B = 1 and A = square wave 		2				
119	2	Which gate is equivalent to bubbled OR gate?	D	1	AND	XOR	NOT	NAND
120	2	A NOT gate has...	A	1	1 input and 1 output	2 input 1 output	1 input 2 output	none of above
121	2	If a 3-input NOR gate has eight input possibilities, how many of those possibilities will result in a HIGH output	B	1	2	1	7	8
122	2	If a signal passing through a gate is inhibited by sending a LOW into one of the inputs, and the output is HIGH, the gate is a(n):	B	1	AND	NAND	NOR	OR
123	2	Implement Boolean expression for Ex-OR gate using NAND gates only.		3				
124	2	The output of a ____ gate is only 1 when all of its inputs are 1.	C	1	OR	NOT	AND	EXOR
125	2	The inputs of a NAND gate are connected together. The resulting circuit is	C	1	OR	AND	NOT	NONE OF ABOVE
126	2	The NOR gate is OR gate followed by	C	1	AND	NAND	NOT	NONE OF ABOVE
127	2	The NAND gate is AND gate followed by	A	1	NOT	OR	NOR	NONE OF ABOVE
128	2	Exclusive-OR (XOR) logic gates can be constructed fromlogic gates.	C	1	OR GATES ONLY	AND gates and NOT gates	AND, OR, NOT Gates	OR gates and NOT gates
129	2	The basic logic gate whose output is the complement of the input is	C	1	OR	AND	INVERTER	NONE OF ABOVE
130	2	The universal gate is	D	1	EXOR	NAND	NOR	BOTH (B) AND (C)
131	2	The NAND gate output will be low if the two inputs are ____.	1&1	1				
132	2	The output of a logic gate is 1 when all its inputs are at logic 0. the gate is either _____.	(a NOR or an EX-NOR)	1				
133	2	_____ AND gates are required to realize $Y = CD+EF+G$	B	1	1	2	3	4
134	2	The following waveform pattern is for ____ 	EXOR GATE	1				
135	2	The following waveform pattern is for ____ 	OR GATE	1				

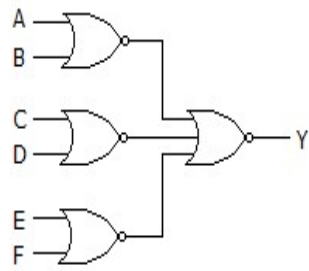
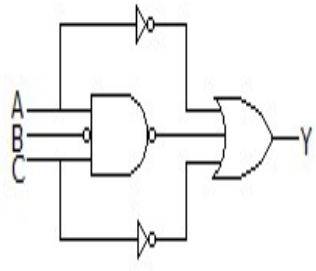
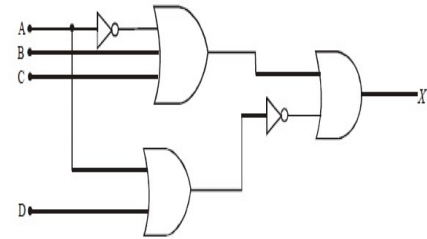
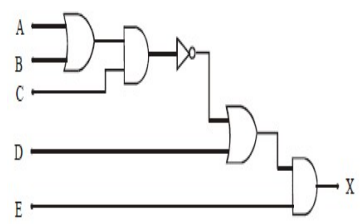
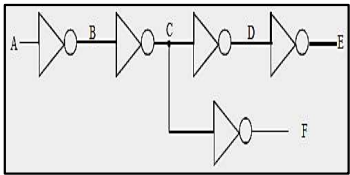
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
136	2	The 8-input XOR circuit shown has an output of $Y = 1$. Correct input combination below (ordered A – H) is _____. 		1				
137	2	Find the logic required at R input. 	R=1	1				
138	2	For a given logic circuit, if A=B=1, and C=D=0. Find output Y 	0	1				
139	2	Show that $A \oplus B = AB' + A'B$ and construct the corresponding logic diagrams using truth table		3				
140	2	Show that $A \odot B = AB + A'B' = (A \oplus B)' = (AB' + A'B)'$ and construct the corresponding logic diagrams using truth table		3				
141	2	Show that $(A+B)(AB)'$ is equivalent to $A \oplus B$ using truth table		3				
142	2	Derive that logic expression that equals to 1 only when the two bit binary numbers A and B have the same value. Draw the logic diagram and construct the truth table to verify the logic.		3				
143	2	Show that $AB + (A+B)'$ is equivalent to $A \odot B$ using truth table		3				
144	2	Find the logical equivalent of the following expression: (a) $A \oplus 0$ (b) $A \oplus 1$ (c) $A \odot 0$ (d) $A \odot 1$ (e) $0 \oplus A'$		5				
145	2	Draw the logic diagram and construct the truth table for following expression: $X = A + B + (CD)'$		3				
146	2	Draw the logic diagram and construct the truth table for following expression: $Y = (AB)(A+B)' + (EF)'$		3				
147	2	Draw the logic diagram and construct the truth table for following expression: $Z = (A'B + CD' + ABC)'$		3				
148	2	A calculator performs the binary addition where the 2's complement binary number 1101100 is obtained with no carry, convert the same into BCD code which is used to display on LCD screen of a calculator.	C	1	11000011	00110100	00100000	00110011
149	2	Excess-3 is also known as _____	D	1	Positive weighted code	Negative weighted code	Cyclic code	Self Complementing Code
150	2	Solve: $(11000111)_{XS-3} = (?)_{GRAY}$		2				
151	2	The specific non-weighted code is used in shaft encoder for the process based instrumentation and data acquisition system that initiates encoder sequence with 01111 to the second sequence as 11111 then evaluate would be the equivalent binary number as per sequence?	A	1	01010 to 10101	01111 to 11111	10000 to 100000	01011 to 10111
152	2	The transmitter transmits the 8-bit of information in form of data packet as D0 (LSB) to D7 (MSB) where D0 defines the odd parity bit, D1 as start bit = '1', D2 to D5 as data bits = $(8)_{2421}$ number and D6 and D7 as stop bits = '01' respectively in a transceiver digital communication system. Evaluate D0	B	1	1	0	01	don'tcare
153	2	Which of the following is true? 1. $(A+B).(AB)' = A \text{ XOR } B$ 2. $AB + (A+B)' = A \text{ XNOR } B$ 3. $A \text{ XOR } B' = A \text{ XNOR } B$	D	1	ONLY 1	ONLY 2	ONLY 3	ALL OF THE ABOVE
154	2	$(000100100011)_{BCD} = (?)_2$	1111011	1				
155	2	Which of the following statements are true? 1. The codes 8421, 2421, 5211, 3321, 4311 are some of the positively weighted codes. 2. The codes 642-3, 631-1, 84-2-1, 74-2-1 are some of the negatively weighted codes. 3. Non-weighted codes do not obey the position-weighting principle.	C	1	Both 1 and 2	both 2 and 3	all are true	all are false
156	2	The specific non-weighted code is used in shaft encoder for the process based instrumentation and data acquisition system that initiates encoder sequence with 10101 to the second sequence as 10111 then evaluate would be the equivalent binary number as per sequence?	C	1	10101 to 10111	00000 to 11111	11001 to 11010	11010 to 11011
157	2	Which of the following statements are true? 1. An XOR gate produces an output 1 only when the inputs are not equal. 2. An XNOR gate produces an output 1 only when the inputs are equal. 3. An XOR is also called anti-coincidence gate. 4. An XNOR is also called coincidence gate.	C	1	Both 1 and 2	both 2 and 3	all are true	all are false

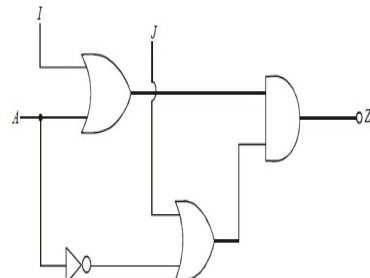
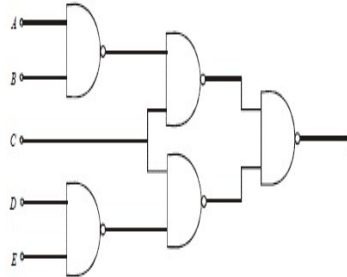
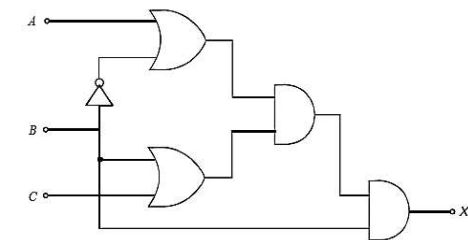
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
158	2	The excess-3 code of decimal 7 is represented by_____.	B	1	0111	1010	0011	0100
159	2	Covert the Gray code 1101 to binary	1001	1				
160	2	Do as directed: 1. Convert the gray code 11011 into decimal. 2. Convert the decimal 2493 into XS-3 code. 3. Convert the decimal 1525 into gray code. 4. $(10101101)_2 = O_{BCD} = O_{GRAY}$		4				
161	2	Convert decimal 86 into BCD, excess-3 and Gray code.		3				
162	2	Convert $(96)_{10}$ to its equivalent Gray code and EX-3 code		4				
163	2	Convert $(10000110)_{BCD}$ to decimal, binary & octal.		3				
164	2	$(396)_{10} = O_{BCD} = O_{GRAY} = O_{XS-3}$		3				
165	2	Decimal 43 in Hexadecimal and BCD number system is respectively	B	1	B2, 0100 0011	2B, 0100 0011	2B, 0011 0100	B2, 0100 0100
166	2	Solve: $(101011)_2 + (001111)_2 = ()_{10} = ()_{16}$		2				
167	2	The 2's complement of the binary number 1001001 in XS-3 is _____		1				
168	2	Given A = P XOR Q, B= P XOR Q' and C= PQ, find output Y = A OR B OR C as per data.		2				
169	2	Solve: $(10100101)_{XS-3} = O_{GRAY}$		2				
170	2	A processor started the operation with reflected code as one bit of change at a time when code is moving from one state to another. The obtained code in one state as 111011 and went to 111001. Convert corresponding code into binary equivalent.		1				
171	2	The transmitter transmits the 8-bit of information in form of data packet as D0 (LSB) to D7 (MSB) where D0 defines the odd parity bit, D1 as start bit ='1', D2 to D5 as data bits = $(F)_{16}$ number and D6 and D7 as stop bits='01' respectively in a transreceiver digital communication system. Evaluate D0	A	1	1	0	01	don'tcare
172	2	Subtract 745.81 from 436.62 given in decimal using 9's complement method.		2				
173	2	Solve: $(101011)_2 - (001111)_2 = (?)_{10} = (?)_{16}$		2				
174	2	The quantity $(337)_{10}$ can be represented in Excess-3 code as:	D	1	100000011001	11000110111	11000111010	11001101010
175	2	Which of the following statements is/are correct for Excess-3 code? 1. It is a self-complementing code. 2. The binary sum of a code and its 1's complement is equal to 1111. 3. It is a weighted code. 4. The binary sum of a code and its 2's complement is equal to 1111. 5. Complement can be generated by inverting each bit pattern.	C	1	all 1, 4, 5	only 1 & 4	all 1, 2, 5	only 2 & 3
176	2	The given logic circuit represents - 	C	1	4-bit binary to decimal converter	4-bit gray to binary converter	4-bit binary to gray converter	4-bit XS-3 to decimal converter
177	2	Unit distance code is the other name of -	B	1	Sequential code	Cyclic Code	Self complementing code	2421 BCD code
178	2	Which of the following number/s is/are in invalid format?	D	1	$(CAFE)_{16}$	$(5208)_8$	$(110000101101)_{XS-3}$	Both B and C
179	2	Which of the following is not an invalid BCD code?	C	1	1011	1010	1001	1100
180	2	An Excess-3 code for the number $(FD)_{16}$ is given by _____	C	1	0010 0101 0011	0011 0010 1000	0101 1000 0110	0101 1001 0110
181	2	Subtract $(3250)_{10}$ from $(72532)_{10}$ using 10's complement method.		2				
182	2	If a 3-input NAND gate has eight input possibilities, how many of those possibilities will result in a HIGH output?	A	1	7	1	3	4
183	2	The 2421 and Excess-3 code of decimal 6 is represented by_____.	A	1	1100 and 1001	1100 and 1100	0110 and 1001	1100 and 0110
184	2	The transmitter transmits the 8-bit of information in form of data packet as D0 (LSB) to D7 (MSB) where D0 defines the even parity bit, D1 as start bit ='1', D2 to D5 as data bits = $(5)_{10}$ in 84-2-1 code and D6 and D7 as stop bits='10' respectively in a transreceiver digital communication system. Evaluate D0.	D	1	8	2	0	1
185	2	Which Logic operation should be performed between A and B for a given output X? 	A	1	XNOR	XOR	OR	NAND
186	2	Substract 75.125 from 84.625 using 12-bit 1's complement arithmetic.		2				
187	2	BCD equivalent code of octal number $(1431)_8$ is	D	1	100101110011	11100000010	100100111001	11110010011
188	2	Convert XS-3 number (11000111) into gray code	D	1	1110011	110110	1100111	1110001
189	2	Subtract $(741.41)_{16}$ from $(436.6)_{16}$ by using decimal (r-1) complement method.		3				
190	2	Perform subtraction of A – B using binary diminished radix complement , where A is octal number (75) & B is octal number (53).		3				
191	2	Do as directed: a) Convert the gray code 11011 into decimal. b) Convert the decimal 2493 into XS-3 code		2				
192	2	Do as directed : a)Convert the decimal 1525 into gray code. b) convert $(10101101)_2$ to 8421 code & gray code		2				
193	2	A processor started the operation with reflected code as one bit of change at a time when code is moving from one state to another. The obtained code in one state as 110001 and went to 101111. Convert corresponding code into binary equivalent?	A	1	100001 to 110101	101110 to 110101	111111 to 110101	0001 to 111111

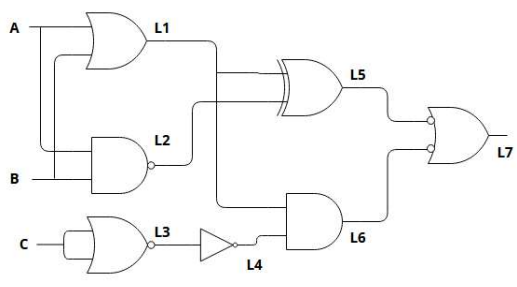
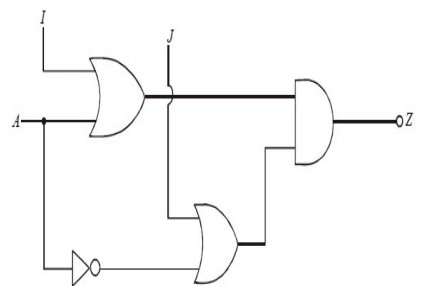
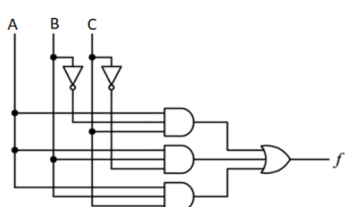
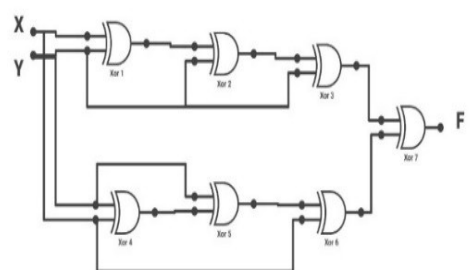
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Digital Electronics (DE) Practice Book (Sem-III 2026)								
Note: This Practice Book is only for reference purpose. LJU Test question paper may not be completely set from Practice Book.								
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
194	2	Convert $(19)_{10}$ into XS-3 code and gray code.	B	1	$(01010000)_{xs-3}$ & 11010	$(01001100)_{xs-3}$ & 11010	$(01001101)_{xs-3}$ & 11000	$(01001110)_{xs-3}$ & 10010
195	2	The decimal equivalent of the excess-3 number 110010100011.01110101 is _____.	C	1	1253.75	861.75	970.42	1132.87
196	2	The 10's complement of $(715)_8$ is _____	D	1	359	953	540	539
197	2	$(98.75)_{10}$ is equivalent to _____	D	1	$(142.6)_8$	$(62.C)_{16}$	$(10011000.01110101)_{BCD}$	All of above
198	2	Assign the proper odd parity bit to the code 111001.	B	1	1111011	1111001	111111	0111011
199	2	Which of the following is false? I. A 1-bit gray code has two code words 0 and 1 representing decimal numbers 0 and 1. respectively. II. 2's complement of a 2's complement of a number is the number itself. III. Excess-3 code for 369 is 011110011100.	C	1	Only 1	Only 2	Only 3	All are false
200	2	The 2's complement of the decimal number $(-541)_{10}$ in hexadecimal is ____.	C	1	DEE	DDD	DE3	DED
201	2	$(1111011)_{GRAY} = ()_{XS-3}$	A	1	10110101	11111111	10111100	10100101
202	2	Which gates are called universal gates? Design exclusive-OR gate having gates using one of the universal gates.		3				
203	2	The transmitter transmits the 8-bit of information in form of data packet as D0 (LSB) to D7 (MSB) where D0 defines the even parity bit, D1 as start bit ='1', D2 to D5 as data bits = (9)10 in 84-2-1 code and D6 and D7 as stop bits='10' respectively in a trans receiver digital communication system. Evaluate D0	B	1	1	0	11	111
204	2	Consider $P = (49.75)_{10}$ and $Q = (55.50)_{10}$. Subtract Q from P using 12-bit 2's complement arithmetic. Express the answer in radix-10 number system.		3				
205	2	Subtract $(77)_{10}$ from $(55)_{10}$ using 8-bit 2's complement arithmetic.		2				
206	2	$(111110011)_{GRAY} = ()_{XS-3}$		2				
207	3	What is the use of boolean identities?	A	1	Minimizing the boolean expression	Maximizing the boolean expression	To evaluate logical identity	Searching of logical expression
208	3	_____ is/are used to implement boolean function.	C	1	Logical notation	Arithmetic logic	Logic gates	expressions
209	3	The boolean function $A + BC$ is a reduced form of _____	B	1	$AB + BC$	$(A + B)(A + C)$	$A'B + AB'C$	$(A + C)B$
210	3	Simplify $Y = AB' + (A' + B)C$.	A	1	$AB' + C$	$AB + AC$	$A'B + AC'$	$AB + A$
211	3	Complement of the expression $A'B + CD'$ is _____	B	1	$(A' + B)(C' + D')$	$(A + B')(C' + D)$	$(A' + B)(C' + D)$	$(A + B')(C + D')$
212	3	$(A + B)(A'B') = ?$	A	1	0	1	AB	AB'
213	3	DeMorgan's theorem states that _____	A	1	$(AB)' = A' + B'$	$(A + B)' = A' * B$	$A' + B' = A'B'$	$(AB)' = A' + B$
214	3	In boolean algebra, the OR operation is performed by which properties?	D	1	Associative properties	Commutative properties	Distributive properties	All of the Mentioned
215	3	The expression for Absorption law is given by _____	A	1	$A + AB = A$	$A + AB = B$	$AB + AA' = A$	$A + B = B + A$
216	3	According to boolean law: $A + 1 = ?$	A	1	1	A	0	A'
217	3	The involution of A is equal to _____	A	1	A	A'	1	0
218	3	$A(A + B) = ?$	D	1	AB	1	0	A
219	3	Find the simplified expression $A'BC' + AC'$.	C	1	B	A+C	$(A+B)C'$	AB
220	3	$(X + Z)(X + XZ') + XY + Y = ?$.	D	1	$XY + Z'$	$Y + XZ' + Y'Z$	$X'Z + Y$	$X + Y$
221	3	$A'(A + BC) + (AC + B'C) = ?$.	D	1	$(AB'C + BC')$	$(A'B + C')$	$(A + BC)$	C
222	3	Simplify the expression $XZ' + (Y + Y'Z) + XY$.	C	1	$(1 + XY')$	$YZ + XY' + Z'$	$(X + Y + Z)$	$XY' + Z'$
223	3	$Y'(X' + Y')(X + X'Y) = ?$	A	1	XY'	$X'Y$	$X + Y$	$X'Y'$
224	3	If an expression is given that $x + x'y'z = x + y'z$, find the minimal expression of the function $F(x,y,z) = x + x'y'z + yz$?	C	1	$y' + z$	$xz + y'$	$x + z$	$x' + y$
225	3	Simplify $XY' + X' + Y'X'$.	C	1	$X' + Y$	XY'	$(XY)'$	$Y' + X$
226	3	Minimize the following Boolean expression using Boolean identities. $F(A,B,C) = (A+BC')(AB'+C)$	D	1	$A + B + C'$	$AC' + B$	$B + AC$	$A(B' + C)$
227	3	Minimization of function $F(A,B,C) = AB(B+C)$ is _____	D	1	AC	B+C	B'	AB
228	3	Algebra of logic is termed as _____	B	1	Numerical logic	Boolean algebra	Arithmetic logic	Boolean number
229	3	Boolean algebra can be used _____	A	1	For designing of the digital computers	In building logic symbols	Circuit theory	Building algebraic functions
230	3	A _____ value is represented by a Boolean expression.	D	1	Positive	Recursive	Negative	Boolean
231	3	$A + AB + ABC + ABCD + ABCDE + =$ _____	B	1	1	A	A+AB	AB
232	3	Simplifying the Boolean expression of $ABC + A'C + AB'C + A'BC$ is _____.	C	1	$A'C + BC$	$A(A+B')$	C	$A'B + AC'$
233	3	Reduce the expression: $A + B(AC + (B + C')D)$		2				
234	3	Reduce the expression: $(A + (BC)')(AB' + ABC)$		3				
235	3	Minimize the Boolean expressions: $X = ((A'B'C')' + (A'B')')'$.		2				
236	3	Find the complement of the following boolean function and reduce to a minimum numbers of literals: $B'D + A'BC' + ACD + A'BC$		3				
237	3	Draw the logic diagram of the following function. Use one OR gate and one AND gate only. $Y = (A+B)(A+C)$		3				
238	3	Given boolean function $F = XY + X'Y' + Y'Z$. Implement it with only OR and NOT gates.		4				
239	3	The logic expression $(A'+B)(A+B')$ can be implemented by giving the inputs A and B to a two input _____	D	1	NOR Gate	NAND Gate	XOR Gate	XNOR Gate
240	3	Which of the given boolean expression is incorrect?	B	1	$A + A'B = A + B$	$A + AB = B$	$(A+B)(A+C)=A+BC$	$(A+B')(A+B)=A$
241	3	The simplified form of boolean expression $(X+Y+XY)(X+Z)$ is	C	1	$X+Y+Z$	$XY+YZ$	$X+YZ$	$XZ+Y$
242	3	$AB + A'C + BC = AB + A'C$ represents which theorem?	A	1	Consensus	Transposition	De Morgan's	None

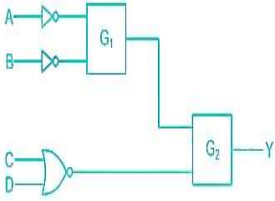
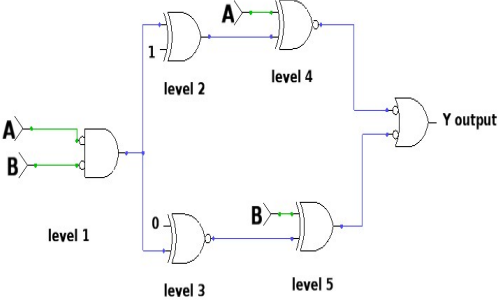
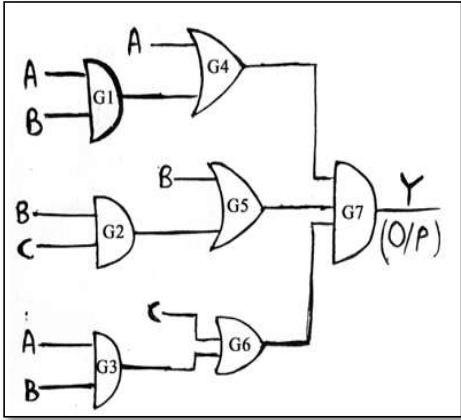
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
243	3	The simplified form of boolean expression $(A'BC+D)(A'D+B'C')$ can be written as	A	1	$A'D+B'C'D$	$AD+BC'D$	$(A'+D)(B'C+D')$	$AD'+BCD'$
244	3	Determine the output X of a logic circuit shown in figure. Simplify the output expression using boolean laws and theorems. Redraw the logic circuit with the simplified expression. 	Simplified Expression is: AB'	3				
245	3	Consider the following boolean expressions: I: $x.y + x'y'$ II: XOR (x', y') III: XOR (x', y) Which of the above expressions represents exclusive NOR operation?	C	1	I and II	I, II and III	I and III	II and III
246	3	Consider the boolean function: $f = (a+bc)(pq+r)$. Complement (f') of a function f is -	B	1	$(a'+b'c')(pq'+r')$	$a'(b'+c')+(p'+q')r'$	$(a'+b'c')+(p'q'+r')$	$a'b'c'+p'q'r'$
247	3	Derive the logic expression for the given logic diagram 	A	1	$C(A+B)DE$	$[C(A+B)D + E']$	$[[C(A+B)D]E']$	ABCDE
248	3	Most Boolean reductions result in an equation in only one form.	A	1	TRUE	FALSE		
249	3	According to property of Commutative law, the order of combining terms does not affect	B	1	initial result of combination	final result of combination	mid-term result of combination	None
250	3	According to boolean algebra which of the following relation is not valid?	D	1	$X(YZ) = (XY)Z$	$X(Y+Z) = XY + XZ$	$X+XZ = X$	$X(X+Y)=1$
251	3	Simplify the following expression: $F = AB'C + AB'C' + ABC$	A	1	$F = AC + AB'$	$F = A'C + AB$	$F = AC' + AB$	$F = AC' + AB'$
252	3	Which of the following options correctly represents the consensus law of Digital Circuits?	A	1	$AB + A'C + BC = AB + A'C$	$A'B + A'C + BC = AB + A'C$	$AB + A'C + BC = AB + (AC)'$	$A'B + A'C + BC = AB + (AC)'$
253	3	Verify the equation by using boolean algebra: $F=AB + AC + BC' = AC + BC'$		4				
254	3	Find out the minimized form of a logical expression $F=(A'B'C' + A'BC' + A'BC + ABC')$.		3				
255	3	The boolean expression $(X+Y)(X+Y')+(X'Y'+X'Y)$ simplifies to _____	A	1	X	Y	XY	X+Y
256	3	Simplify the following boolean expression to four literals: $F = A'C + C'D + B'C + AB$	$AB+C+D$	3				
257	3	The reduced form of complement of equation $f = [(ab)'a][(ab)'b]$	B	1	0	1	$a'b+ab'$	$a'b'$
258	3	The reduced form of dual of equation $f = [(ab)'a][(ab)'b]$	B	1	0	1	$a'b+ab'$	$a'b'$
259	3	Which of the given boolean expression is incorrect?	D	1	$(A'+B')' = AB$	$(A+ (B'))' = A' B'$	$(A'B')' = A+B$	$(A'B'')' = A'+B$
260	3	The boolean equation $x = [(A+B')(B+C)]B$ can be simplified to -	C	1	$X = A'B$	$X = AB'$	$X = AB$	$X = A'B'$
261	3	Find the output boolean function for the given logic circuit. 	A	1	$X = A'BC$	$X = AB'C$	$X = ABC$	$X = A'B'C$
262	3	The output Y of the logic circuit is - 	A	1	1	0	X	X'
263	3	Simplify the following boolean expression and realize it using NOR gates only: $Y = A\bar{B} + AB\bar{C} + ABCD + ABC\bar{D}$		3				
264	3	Simplify the following boolean expression and realize it using NAND gates only: $ABC[AB + \bar{C}(BC + AC)]$		3				
265	3	Match the appropriate pairs: (1) $x+(y+z)=(x+y)+z$ (A) Commutative Law (2) $x+y=y+x$ (B) Absorption Law (3) $x+xy=x$ (C) Complementation law (4) $(x')'=x$ (D) Associative Law	C	1	1-B, 2-A, 3-D, 4-C	1-D, 2-A, 3-C, 4-B	1-D, 2-A, 3-B, 4-C	1-C, 2-A, 3-D, 4-B
266	3	Simplify: $f = AB + ABC + \bar{A}B + A\bar{B}C$	$B+AC$	2				

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D																																				
267	3	Minimize the following function using boolean algebra: $f = \bar{A}BCD + AB\bar{C}\bar{D} + AB\bar{C}D + ABCD + ABC\bar{D} + A\bar{B}\bar{C}D + A\bar{B}CD + A\bar{B}C\bar{D}$		2																																								
268	3	Simplify: $f = (A + \bar{B}C) + \overline{(A + \bar{B}C)} = 1$		3																																								
269	3	Simplify the following boolean expression which represents the output of a logical decision circuit $f(w, x, y, z) = x + xyz + \bar{x}yz + wx + \bar{w}x + \bar{x}y$	x+y	2																																								
270	3	Simplify the following boolean expression which represents the output of a logical decision circuit $f(A, B, C, D, E) = (AB + C + D)(\bar{C} + D)(\bar{C} + D + E)$	ABC'+D	3																																								
271	3	Prove the following identity: $\overline{\overline{AB}} + \bar{A} + AB = 0$		2																																								
272	3	Simplify the following function by using boolean algebra: $Y = AB'C'D + A'B'D + BCD' + A'B + BC'$		3																																								
273	3	Simplify the following function by using boolean algebra: $Y = (AB + A'C + BC)(A+B'+AB')$	AB+A'B'C	2																																								
274	3	Construct a logic circuit to give an output without any reduction in number of gates. Give the logic gates output step by step. $X = (\overline{AB} + \bar{A}C)(\bar{A}D + C)$		3																																								
275	3	Find out the complement of boolean expression : AB (B'C+AC)		2																																								
276	3	The boolean function $Y = AB + CD$ is to be realized using only 2-input NAND gates. The minimum number of NAND gates required is -	B	2	2	3	4	5																																				
277	3	Realize the given logic circuit into appropriate boolean expression and also minimize it. 		3																																								
278	3	Make a suitable logic expression from the given truth table <table border="1" data-bbox="260 1421 564 1733"><thead><tr><th>A</th><th>B</th><th>C</th><th>Y</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></tbody></table>	A	B	C	Y	0	0	0	1	0	0	1	1	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	0	1	1	1	0		2				
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279	3	_____ NAND gates required to implement the function : $F = (\bar{X} + \bar{Y})(Z + W)$		3																																								
280	3	The boolean expression for the given circuit is _____ 	A	1	A {F + (B + C) (D + E)}	A [F + (B + C) (DE)]	A + F + [(B + C) (D + E)]	A [F + (BC) (DE)]																																				
281	3	DeMorgan's first theorem shows the equivalence of _____	B	1	OR gate and Exclusive OR gate	NOR gate and Bubbled AND gate	NOR gate and NAND gate	NAND gate and NOT gate																																				
282	3	$((AB)' + (AC))' =$ _____	B	1	A+B+C	ABC	A'BC	(A+B+C)'																																				
283	3	Consider four distinct input boolean variables, in which first two and last two variables are ANDed. Then after output of both operation is Ored together. Implement the circuit based on the given information and also find its boolean expression.		2																																								
284	3	What is the boolean expression for Y in above circuit ? 		2																																								

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
285	3	Find out the simplified boolean equation for the given logic circuit. 	A	1	$(A + B)(C + D)(E + F)$	$A+B+C+D+E+F$	ABCDEF	$ABC + DEF$
286	3	Four inputs A, B, C, D are fed to a NOR gate. The output of NOR gate is fed to the two successive inverter. The final output is given by - _____	B	1	$A+B+C+D$	$(A+B+C+D)'$	ABCD	$(ABCD)'$
287	3	For the logic circuit of the given figure, the minimized expression is - 	A	3	$Y = (AB'C)'$	$Y = A+B+C$	$Y = A + B$	$Y = ABC$
288	3	Choose the correct statement for 'literal' - (I) Literal is a primed variable only. (II) The minimization of the number of literals and the number of terms results in a circuit with less equipment (III) We can not reduce the number of literals in any given Boolean function. (IV) In the Boolean equation $y=a'b+b'c+c'd$, the terms $a'b$, $b'c$ and $c'd$ are known as literals.	B	1	Only I and III	Only II	Only II and IV	All are correct
289	3	Simplify the boolean expression : $y(wz'+wz)+xy$		2				
290	3	Reduce the following boolean function: $ABC + A'B'C + A'BC + ABC' + A'B'C'$		2				
291	3	Reduce the following boolean function: $X = \bar{B}D + \bar{A}B\bar{C} + ACD + \bar{A}BC$		2				
292	3	Find the complement of $F = x + yz$, then show that $F \cdot F' = 0$ and $F + F' = 1$		2				
293	3	Find the complement of the following Boolean functions and reduce them to a minimum number of literals: $(A' + C)(A' + C')(A + B + C'D)'$		3				
294	3	Write the Boolean expression for output X for the logic circuit shown in figure. Also show the output of each stage of logic gates: 		3				
295	3	Write the Boolean expression for output X for the logic circuit shown in figure. Also show the output of each stage of logic gates: 		3				
296	3	Construct a logic circuit using INVERTER, AND and OR gates for the Boolean expression $X = \overline{(A+B+\bar{C}D\bar{E})} + \bar{B}C\bar{D}$		3				
297	3	A network of cascaded inverters shown in fig. If a logic 1 is applied to A, Determine the logic levels from point B through F. 		2				
298	3	Using boolean laws and rules, simplify the expression: $Z = A'BC + AB'C' + A'B'C' + AB'C + ABC$		3				
299	3	Using boolean laws and rules, simplify the expression: $Z = (AB+AC)' + A'B'C$ and prove that $Z = A' + B'C'$		3				

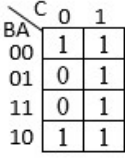
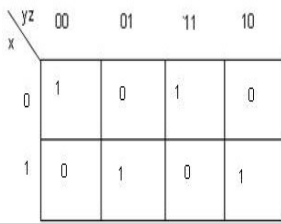
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D						
300	3	Demorganize the expression: $\overline{(A + B)\bar{C}\bar{D} + (E + \bar{F})}$		2										
301	3	Simplify the boolean expression: $T[x,y,z] = (x+y) (x'(y'+z'))'+x'y'+x'z'$.		3										
302	3	The circuit shown in Fig. is used to implement the function $Z = f(A, B) = A + B$. What values should be selected for I and J ? 		3										
303	3	Determine the logic function realized by the circuit shown in Fig. Also minimize the output function f using laws boolean algebra. 		3										
304	3	Determine by means of truth table, the validity of De Morgan's theorem for three variables: $(ABC)' = A' + B' + C'$		3										
305	3	A switch board contains four electrical switches which are connected in a sequence in a room: 1st switch as main power supply switch of the room, 2nd switch belongs to a fan, 3rd switch belongs to a TV and 4th switch belongs to 0 W bulb. The main power supply switch is also directly connected to the 0W bulb. Design the boolean expression of the switch board of the room such that user can operate switch board as per his requirement. Also show how the boolean expression can be minimized.		3										
306	3	Boolean arithmetic is a :	A	1	way to express logic statements in a traditional mathematical equation format	terrible fraud perpetrated by philosophers to disprove things they don't agree with	very difficult calculation used in astronomy	fast way to solve problems around the house						
307	3	Determine and minimize the Boolean expression for the output, X of a logic circuit shown in figure. Also give your comment on the minimized output. Which logic gate is represented by simplified output? 	AND gate	3										
308	3	Simplify the following boolean expression using laws of Boolean algebra. (1) $f(w,x,y,z)=x+x'y+z+x'yz+wx+w'x+x'y$ (2) $Y=AB'+ABC'+ABCD+ABCD'$		3										
309	3	As per _____ theorem, $PQ+P'R+QR =$ _____	C	1	Transposition, $PQ+QR$	Absorption, $PQ+P'R$	Consensus, $PQ+P'R$	Consensus, $PQ+PR'$						
310	3	Which of the following statements are true? 1. An XOR gate produces an output 1 only when the inputs are not equal. 2. An XNOR gate produces an output 1 only when the inputs are equal. 3. An XOR is also called anti-coincidence gate. 4. An XNOR is also called coincidence gate.	D	1	Both 1 and 2	Both 3 and 4	Both 2 and 3	All are true						
311	3	Evaluate which one is correct? <table border="1"><tr><td>(1)</td><td>$(A'+B') \oplus B' = AB'$</td></tr><tr><td>(2)</td><td>$(AB)' \odot (A+B) = 1$</td></tr><tr><td>(3)</td><td>$A' \oplus A \odot 1 = A'+A$</td></tr></table>	(1)	$(A'+B') \oplus B' = AB'$	(2)	$(AB)' \odot (A+B) = 1$	(3)	$A' \oplus A \odot 1 = A'+A$	C	1	Only 1	Only 2	Only 3	Only 4
(1)	$(A'+B') \oplus B' = AB'$													
(2)	$(AB)' \odot (A+B) = 1$													
(3)	$A' \oplus A \odot 1 = A'+A$													
312	3	If a 3-input NAND gate has eight input possibilities, how many of those possibilities will result in a LOW output?	A	1	1	7	5	6						

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Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
313	3	The given logic diagram describes the process of an industrial operation that depends on distinct values of A, B and C. Find the boolean expressions of each level (Level 1 to level 7) also define following logic diagram belongs to which equivalent logic gate? 		4				
314	3	Consider the following boolean expressions and Check whether the given statement/s is/are Correct or not. I: $MN' + M' + M'N' = (MN)'$ II: $Q' (P' + Q') (P + P'Q) = PQ'$	C	1	Only II and IV are correct	Only III is correct	Only I, II and III are correct	All are correct
315	3	The circuit shown in Fig. is used to implement the function $Z = f(A, B) = (AB)'$. What values should be selected for I and J ? 	Both A & B	1	I= A' AND J= B'	I= 1 AND J=B'	I=A AND J= B	I= B' AND J= A'
316	3	Given $W = [(A \text{ XOR } B) \text{ XNOR } 1]$, $X = [(A \text{ XOR } B') \text{ XOR } 0]$, $Y = [(A \text{ XNOR } B) \text{ XOR } 1]$ and $Z = [(A' \text{ XOR } B) \text{ XNOR } 0]$. Find Output Function = $W \text{ OR } X \text{ OR } Y \text{ OR } Z$ as per data and simplify it using basic theorems and construct simplified function with basic logic gates.		2				
317	3	Determine the output F of a logic circuit shown in figure with truth table and logical expression. Simplify the output expression using boolean laws and theorems. Redraw the logic circuit of simplified expression with two gates only. 		3				
318	3	If a signal passing through a gate is inhibited by sending a LOW into one of the inputs, and the output is HIGH, the gate is a(n):	B	1	AND	NAND	OR	NOR
319	3	Choose correct statement/s from following i) an inverter performs complementation operation ii) process performed by inverter is known as inversion iii) an inverter contains two or more input terminal and one output terminal.	A	1	Only I & II	Only I & III	Only II & III	All of the above
320	3	if the boolean expression $P'Q + QR + PR$ is minimized , the expression becomes	B	1	$P'Q + QR$	$P'Q + PR$	$QR + PR$	$P'Q + QR + PR$
321	3	$A + AB = A$; $A(A+B) = A$ represents which law ?	B	1	commutative	Absorption	Consensus	Transposition
322	3	Exclusive-OR (XOR) logic gates can be constructed fromlogic gates	D	1	AND, OR, NOT	OR Gate & NOT gate	NOR & NOT Gate	a & c both
323	3	The circuit shown below generates the function of $F =$ _____ 	A	1	$x \oplus y$	0	$x\bar{y} + yx + \bar{y}x$	$x + y$
324	3	Transmitter transmit 8 bit of information in form of data packet D0 (LSB) to D7 (MSB) where D0 Defines EVEN PARITY BIT, D1 as start bit '1' , D2 to D5 as data bits (E)16 number & D6 & D7 as stop bit '10' respectively in transreceiver digital communication system. evaluate the D0 = _____	A	1	1	11	10	0

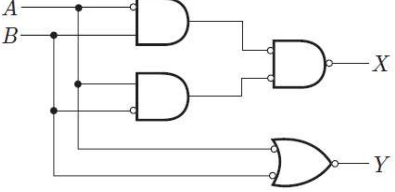
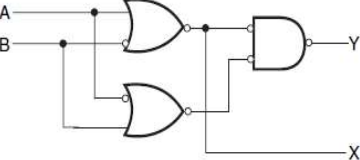
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
325	3	In the figure shown, the output Y is required to be $Y = A.B + C'.D'$. The gates G1 and G2 must be respectively 	D	1	AND & OR	OR & NAND	NAND & OR	NOR & OR
326	3	Draw the logic diagram of boolean expression $(A+B)(CD)(E+F)'G'$ & its complement separately using only two input basic logic gates (do not use NAND , NOR , X-OR , X-NOR logic gates)		3				
327	3	Which of the following is true? 1. $(A+B).(AB)' = A \text{ XOR } B$ 2. $AB+(A+B)' = A \text{ XNOR } B$ 3. $A \text{ XOR } B' = A \text{ XNOR } B$	D	1	Both 1 and 2	Both 1 and 3	Both 2 and 3	All are true
328	3	Simplify the Boolean expression. $AB+(AC)'+AB'C(AB+C)$		2				
329	3	The given logic diagram depends on values of A & B. Find the Boolean expressions of each level till output Y and also define following logic diagram belongs to which equivalent gate? 		4				
330	3	Match the following 1) $A+A'$ P. 0 2) $A \oplus A$ Q. 1 3) $0 \oplus A$ R. A 4) $A'A'$ S. A'	B	1	1-P, 2-Q, 3-R, 4-S	1-Q, 2-P, 3-R, 4-S	1-S, 2-R, 3-P, 4-Q	1-Q, 2-P, 3-S, 4-R
331	3	$MN(M + N') + M(N + N') = \underline{\hspace{2cm}}$	A	1	M	N	M'	N'
332	3	Find out the Boolean Expression for Logic Diagram given below and simplify the output in the minimal expression, also implement the simplified expression using the AOI logic. 		4				
333	3	Match the following with correct logic expression: Column A Column B 1. $A' \odot A \oplus A'$ w. 1 2. $A' \oplus 0 \oplus A$ x. A' 3. $A \oplus A' \odot A$ y. 0 4. $A' \odot 1 \odot A$ z. A	B	1	1-x, 2-z, 3-w, 4-y	1-x, 2-w, 3-z, 4-y	1-w, 2-z, 3-x, 4-y	1-y, 2-w, 3-z, 4-x
334	4	The Sum of all Minterm is _____	B	1	0	1	2	4
335	4	The Product of all Maxterm is _____	A	1	0	1	2	4
336	4	The minterm related to $F(w,x,y,z) = m_6$ is	A	1	$w'xyz'$	$wxyz$	$wx'y'z$	$wx'yz$
337	4	The Maxterm related to $F(w,x,y,z) = M_7$ is	A	1	$w+x'+y'+z'$	$w'+x+y+z$	$w+x+y+z$	$w'+x+y'+z$
338	4	Convert SOP into POS : $F(A,B,C) = \sum m(1,3,7)$	D	1	$\pi M(0,2,4,6)$	$\pi M(2,4,5,6)$	$\pi M(0,2,4)$	$\pi M(0,2,4,5,6)$
339	4	Convert POS into SOP: $F(w,x,y,z) = \pi M(0,1,2,6,10,12,14,15)$	C	1	$\sum m(3,4,5,7,8,9,11,13,16)$	$\sum m(0,3,4,5,7,8,9,11,13)$	$\sum m(3,4,5,7,8,9,11,13)$	$\sum m(3,4,5,8,9,11,13)$
340	4	Express the boolean function $F = A + B'C$ in sum of minterm		2				
341	4	Express the Boolean function $F = AB + A'C$ in a product of maxterm.		2				
342	4	Convert into Product-of-Maxterms: $A(A'+B)(C')$		2				
343	4	Convert into Sum-of-Minterms: $A' + B + CA$		2				
344	4	Express Function in Product of Maxterms $F(x,y,z) = (xy + z)(y + xz)$		2				
345	4	Express the Boolean function in sum of minterms $F(A,B,C) = (A'+B)(B'+C)$		2				
346	4	Express the Boolean function in sum of minterms $F(A,B,C,D) = D(A'+B) + B'D$		2				
347	4	Express the Boolean function in sum of minterms and product of max terms. $F(A,B,C,D) = (A+B'+C)(A+B')(A+C'+D')(A'+B+C+D')(B+C'+D')$		3				

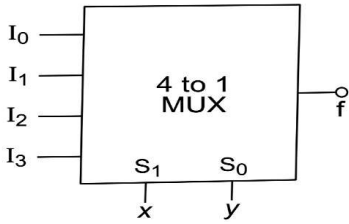
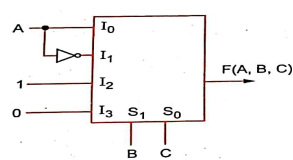
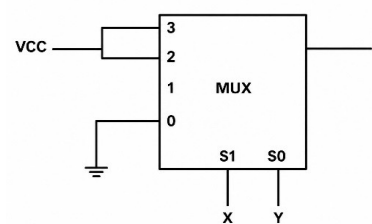
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Sr. No.	unit_num ber	question_text	answer_text	marks	option A	option B	option C	option D
348	4	Express the Boolean function in sum of minterms and product of max terms. $F(X,Y,Z) = (XY+Z)(Y+XZ)$		3				
349	4	Express the Boolean function in sum of minterms and product of max terms. $F(X,Y,Z) = 1$		3				
350	4	Express the Boolean function in sum of minterms and product of max terms. $F(W,X,Y,Z) = Y'Z+WXY'+WXZ'+W'X'Z$		3				
351	4	Expand A'+ B' to minterm and maxterm		2				
352	4	Expand A+BC'+ABD'+ABCD to minterm and maxterm		2				
353	4	Convert the following Boolean function in product of maxterms and sum of minterms: $F(m,n,o,p) = n'o'p+nop+mop'+m'n'o+m'no'p$		3				
354	4	Expand $A(A'+B)(A'+B+C')$ to minterm and maxterm		2				
355	4	Convert the following to other canonical form $F(x,y,z) = \sum(1,3,7)$		2				
356	4	Convert the following to other canonical form $F(A,B,C,D) = \sum(0,2,6,11,13,14)$		2				
357	4	Convert the following to other canonical form $F(X,Y,Z) = \pi(0,3,6,7)$		2				
358	4	Convert the following to other canonical form $F(A,B,C,D) = \pi(0,1,2,3,4,6,12)$		2				
359	4	A Karnaugh map is an abstract form of _____ diagram organized as a matrix of squares	A	1	Venn	Cycle	Point	Block
360	4	Which of the following code is employed by K-Map for simplification of Boolean Expression?	B	1	XS-3	Gray	BCD	Parity
361	4	The 3-variable Karnaugh Map (K-Map) has _____ cells for min or max terms	C	1	3	6	8	2
362	4	Which of the following is NOT considered for forming groups in K-map?	A	1	Diagonal	Rolling	Vertical	Horizontal
363	4	There are 3 Variable in the boolean function and the value of the function is 1. Find the number of cells in the K-Map which will contain a 1 when SOP expression is used	B	1	3	8	1	0
364	4	There are 3 Variable in the boolean function and the value of the function is 1. Find the number of cells in the K-Map which will contain a 0 when SOP expression is used	D	1	3	8	1	0
365	4	Given $F(A, B, C, D) = \sum m(0, 1, 2, 6, 8, 9, 10, 11) + \sum d(3, 7, 14, 15)$ is a Boolean function, where m represents min-terms and d represents don't cares. The minimized logical fuction F is	B	1	Independent of variables	Depedent on 2 variable	Depedent on 3 variable	Depedent on 4 variable
366	4	In simplification of a Boolean function of n variables, a group of 2^m adjacent 1s leads to a term with	D	1	$m - 1$ literals less than the total number of variables	$m + 1$ literals less than the total number of variables	$n + m$ literals	$n - m$ literals
367	4	Looping on a K-map always results in the elimination of _____	C	1	Variables within the loop that appear only in their complemented form	Variables that remain unchanged within the loop	Variables within the loop that appear in both complemented and uncomplemented form	Variables within the loop that appear only in their uncomplemented form
368	4	Digital input signals A, B, C with A as the MSB and C as the LSB are used to realize the Boolean function $F = m_0 + m_2 + m_3 + m_5 + m_7$, where m_i denotes the i th minterm. In addition, F has a don't care for $m_1 + m_4 + m_6$. The simplified expression for F is given by:	A	1	1	$(A + C)(A' + C')$	$A'C' + AC$	$A'C + AC'$
369	4	Simplify the Boolean function using K-Map: $F(w,x,y,z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$		3				
370	4	Simplify the Boolean function using K-Map: $F(w,x,y) = \sum(0,1,3,4,5,7)$		2				
371	4	Obtain the simplified expression using K-Map in sum of product for the Boolean functions $F = \sum(0,1,4,5,10,11,12,14)$.		3				
372	4	Simplify the Boolean function $F(x,y,z) = \sum m(0,1,2,3,4,5,6)$ using K- map.		3				
373	4	Simplify the following Boolean function using K-map $F(w,x,y,z) = \sum(1, 3, 7, 11, 15)$		3				
374	4	Simplify the following Boolean function using K-Map. $F = A'B'C' + B'CD' + A'BCD' + AB'C$.		4				
375	4	Obtain the simplified expressions in sum of products using K-map: $F = x'z + w'xy' + w(x'y + xy')$		4				
376	4	Simplify following boolean function using K-Map: $Y = \sum m(1, 5, 7, 9, 11, 13, 15)$		3				
377	4	Simplify following boolean function using K-Map: $Y = \sum m(0, 2, 3, 5)$		2				
378	4	Simplify following boolean function using K-Map: $Y = \sum m(1, 3, 5, 9, 11, 13)$		3				
379	4	Simplify following boolean function using K-Map: $Y = \sum m(0, 2, 5, 6, 7, 8, 10, 13, 15)$		3				
380	4	Simplify following boolean function using K-Map: $Y = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$		3				
381	4	Simplify following boolean function using K-Map: $Y = \sum m(1, 3, 4, 5, 7, 9, 11, 13, 15)$		3				
382	4	Simplify Boolean function using K-Map: $F(w,x,y,z) = \sum(1,3,5,8,9,11,15)$		3				
383	4	Simplify the Boolean Function with Karnaugh map: $F(w,x,y,z) = \sum(0,1,2,4,5,6,8,9,12,13,14,15)$		3				
384	4	Simplify the Boolean Function with Karnaugh map: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$		4				
385	4	Simplify the Boolean function $F(x,y,z) = \sum(0,2,4,5,6)$ using K- map.Explain groups.		2				
386	4	Simplify following boolean function using K-Map: $Y = AB'C'D' + AB'C'D + AB'CD + AB'CD'$		4				
387	4	Simplify using K-map $F = AB'C + AB'C'D + ABC'D + ABC$		4				
388	4	Simplify using K-map $F = A'B'C + A'BC + ABC + ABC'$		3				
389	4	Minimize following Boolean function using K-map: $X(A,B,C,D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 15)$		3				
390	4	Simplify the Boolean function, $F = \sum m(0,1,2,5,8,9,10)$		3				
391	4	Minimize following Boolean function using K-map $F = \sum m(1, 2, 4, 6, 7, 11, 15) + \sum d(0, 3)$		4				
392	4	Minimize the following function using K-Map $F = \sum m(0,2,6,10,11,12,13) + d(3,4,5,14,15)$		4				
393	4	Simply the Boolean Function using K-map : $F(W,X,Y,Z) = \sum(1, 3, 7, 11, 15)$ with don't care conditions $d(W,X,Y,Z) = \sum(0, 2, 5)$		4				
394	4	Reduce the given function using K-map $F(A,B,C,D) = \sum m(0,1,3,7,11,15) + \sum d(2,4)$		4				

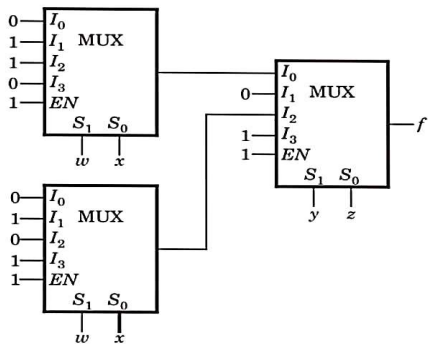
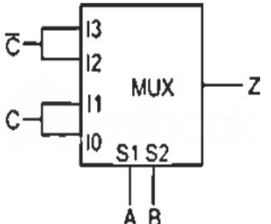
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Sr. No.	unit_num ber	question_text	answer_text	marks	option A	option B	option C	option D
395	4	Minimize the following logic function using K-maps F(A,B,C,D) = $\sum m(1,3,5,8,9,11,15) + d(2,13)$		4				
396	4	Minimize the following function using K-map $F(w,x,y,z) = \sum m(0,1,2,3,6,7,13,14) + \sum d(8,9,10,12)$		4				
397	4	Minimize the following functions using K Map $F = \prod M(0,4,9,10,11,14,15)$		3				
398	4	Minimize the logic function $F(A,B,C,D) = \pi M(1, 2, 3, 8, 9, 10, 11, 14) . d(7, 15)$ Use Karnaugh map. Draw the logic circuit for simplified function using NOR gates only.		4				
399	4	Simplify Boolean function $F(w,x,y,z) = \sum (0,1,2,4,5,6,8,9,12,13,14)$ using K-map and Implement it using NAND gates only		4				
400	4	Solve the following Boolean functions by using K-Map. Implement the simplified function by using logic gates $F = (w,x,y,z)$ $= \sum (0,1,4,5,6,8,9,10,12,13,14)$		4				
401	4	Reduce the expression $F = \sum m(0,2,3,4,5,6)$ using K-map and implement using NAND gates only.		4				
402	4	Implement the function $F = \sum (0,6)$ with NAND gates only.		3				
403	4	Implement the function $F = \sum (0,6)$ with NOR gates only		3				
404	4	Obtain the simplified expressions in SOP for the following Boolean function using K-Map method. And implement it using NAND gate. $F(A,B,C,D) = ABC + AB'C + BCD' + A'CD$		4				
405	4	Reduce using mapping the expression $\sum m(0, 1, 2, 3, 5, 7, 8, 9, 10, 12, 13)$ and implement it with any universal logic.		4				
406	4	Reduce using mapping the expression $\pi M(2, 8, 9, 10, 1, 12, 14)$ and implement it with any universal logic		4				
407	4	Reduce using mapping the expression $\sum m(1, 5, 6, 12,13, 14) + d(2,4)$ and implement it in universal logic.		4				
408	4	Reduce using mapping the expression $\sum m(0,1, 3, 5, 6, 12,13, 14) + d(2,7,8,15)$ and implement it in universal logic.		4				
409	4	Reduce using mapping the expression $\sum m(0, 1, 4, 7, 13, 14) + d(5, 8, 15)$ and implement it in universal logic.		4				
410	4	Reduce using mapping the expression $\pi M(0,1 ,3 4, 5, 7, 10, 13, 14, 15)$ and implement it in universal logic.		4				
411	4	Reduce using mapping the expression $\pi M(2, 4, 6, 8, 10, 12, 15)$ and implement it in universal logic.		4				
412	4	Reduce using mapping the expression $\pi M(0, 1, 4, 6, 8, 9, 11)$ and $d(2, 7, 13)$ and implement it in universal logic.		4				
413	4	Reduce using mapping the expression $\pi M(2, ,4 ,5 7, 9, 12)$ and $d(0, 1, 6)$ and implement it in universal logic.		4				
414	4	The inputs to a computer circuit are the 4 bits of the binary number A3A2A1A0 The circuit is required to produce a 1 if any of the following conditions hold. 1. The MSB is 1 and all other bits are 0. 2. A2 is a 1 and the all other bits are 0. 3. All of the 4 bits are 0. 4. A1 and A0 is 1 and other bits are 0 Obtain a minimal expression using K-map and implement it with NAND gate		5				
415	4	A8A4A2A1 is an Binary input to a logic circuit whose output is a 1 when A8=0, A4=0 and A2=1,or when A8=0 and A4=1.Design the simplest possible logic circuit with NAND gate only using K-map in SOP form.		5				
416	4	The _____ is a very useful and convenient tool for simplification of Boolean functions for large numbers of variables.	A	1	Quine McCluskey tabulation method	K-map	SOP K-map	POS k-map
417	4	The implicants which will definitely occur in the final expression are called	C	1	Prime implicant	Non-prime implicant	Essential Prime implicant	implicant
418	4	_____ is a group of minterms which can't be combined with any other groups.	C	1	Odd implicant	Even implicant	Prime implicant	Non-prime implicant
419	4	In the Quine-McClusky method of minimization of the function f(A, B, C, D) the PI corresponding to -1 1- is	A	1	BC	AD	B'C'	A'D'
420	4	In the Quine-McClusky method of minimization of the function f(A, B, C, D) the PI corresponding to -1 -0 is	A	1	BD'	B'D	AC	A'C'
421	4	Unchecked terms in the PI table forms are called _____	A	1	Prime implicant	Non-prime implicant	Essential Prime implicant	implicant
422	4	Simplify following Boolean function by using the tabulation method $F = \sum (0,1,3,7,8,9,11,15)$		5				
423	4	Simplify the following Boolean expression by means of the Tabulation method. $F(A,B,C,D) = \sum m(1,2,3,5,6,7,8,9,12,13,15)$.		5				
424	4	Simplify the following boolean function using tabulation method and draw logic diagram using AND-OR-NOT gates. $F(w, x, y, z) = \sum (0, 1, 2, 8, 10, 11, 14, 15)$		5				
425	4	Obtain the set of prime implicants for $\sum m(0, 1, 6, 7, 8, 9, 13, 14, 15)$		5				
426	4	Obtain the set of prime implicants for $\sum m(0,1 ,3, 4, 5, 7, 10, 11, 13, 14, 15)$		5				
427	4	Obtain the set of prime implicants for $\sum m(0,1, 2 ,3 6, 7, 8, 10, 11, 12, 15)$		5				
428	4	Obtain the set of prime implicants for $\sum m(5, 6, 12, 13, 14)$		4				
429	4	Simplify Boolean function by using the tabulation method $F = \sum m(1,2 ,5 6, 8, 9, 10,11,12,15)$		5				
430	4	Simplify Boolean function by using the tabulation method $F = \sum m(0, 1, 2,4, 6,7, 8, 9, 11,13)$		5				
431	4	Simplify Boolean function by using the tabulation method $F = \sum m(0,1,2,4,5,6, 7,9,12)$		5				
432	4	Find Essential Prime Implicants of $F = \sum m(1,2,3,5,7,10,11,14,15)$		5				
433	4	Design Prime Implicant chart and calculate essential Prime Implicants for the function $F = \sum m(0,1,4,5,7,8,13,14,15)$		5				
434	4	Calculate number of essential prime implicants for $F = \sum m(3,4,5,9,10,11,14)$		4				
435	4	Simplify the following Boolean function $F(W,X,Y,Z) = \sum m(2,6,8,9,10,11,14,15)$ using Quine-McClukey tabular method.		5				
436	4	Using the tabular method of simplification, find all equally minimal solutions for the function $F(A,B,C,D) = (1,4,5,10,12,14)$		5				
437	4	Obtain the simplified Boolean expression for the following Boolean function with don't care condition and implement the simplified function with NOR gates only. $F(w, x, y, z) = w\bar{y}\bar{x}\bar{z} + w\bar{x}\bar{y}z + x\bar{y}z\bar{w} + yzwx + wy\bar{z}\bar{x} + wzxy + ywxz + wxyz$ $d(w, x, y, z) = w\bar{x}\bar{y}z + z\bar{w}xy + y\bar{z}wx + y\bar{z}wx$		5				

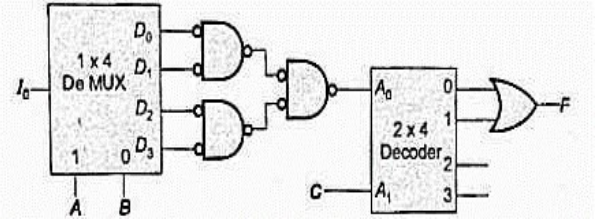
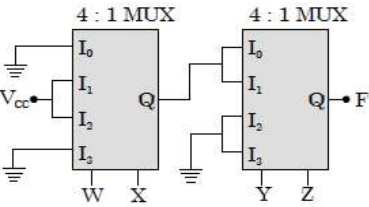
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438	4	Convert the following Boolean function in product of maxterms and sum of minterms: $F(m,n,o,p)=n\bar{o}\bar{p}+nop+m\bar{o}+m\bar{n}\bar{o}+m\bar{n}\bar{o}\bar{p}$		3				
439	4	Consider the function $f(A, B, C, D) = (0,1,2,3,5,7,8,10,12,13,15)$ and its prime implicants are A'B' from pair (0,1,2,3) , B'D' from (0,2,8,10) , A'D from		3				
440	4	Calculate essential prime implicants using Prime implicant chart and also obtain simplified logical expression for $F = (3,4,5,6,7,10,11,12,13,15)$		4				
441	4	Simplify the following boolean function $F(w, x, y, z) = \sum m(6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$ using Quine-McCluskey method.		4				
442	4	Which Boolean function the following Karnaugh map represent? 	B	1	A + C	A' + C	A + C'	A + CA'
443	4	$\bar{A}B + CD$ is the simplified version of the Boolean expression F only if there were a ‘don’t care’ entry. What is that don’t care term ? $F = ABCD + \bar{A}\bar{B}CD + \bar{A}B$	A	1	AB'CD	AB'C'D	A'BCD	A'BC'D
444	4	There are 4 variables P, Q, R and S in Boolean function and the value of the function is 0. Find the number of cells in the K-map which will contain 0 when POS expression is used.	D	1	15	1	0	16
445	4	Convert the following Boolean function in product of max terms and sum of minterms: $F(p,q,r,s) = (p' + q + r) (q' + r + s) (p + s')$		2				
446	4	Obtain the simplified Boolean expression for the following Boolean function with don’t care condition and implement the simplified function with NOR gates only. $F(w, x, y, z) = w'y'x'z' + w'x'yz' + x'yz'w + yzwx' + wy'z'x + wzxy' + ywzx' + wxyz$ $d(w, x, y, z) = w'x'yz + zw'xy + y'z'wx' + y'zwx'$		5				
447	4	Obtain the simplified boolean expression for the following boolean function with don’t care condition and implement the simplified function with two input NAND gates only. $F(w,x,y,z) = w' x z + w' y z + x' z' y + w y' x z$ $d = w y z$		5				
448	4	Simplify Boolean function $F(W, X, Y, Z) = \sum (0, 1, 2, 4, 5, 6, 8, 9, 10, 12, 13)$ using K-map and implement it using NAND gates only.		3				
449	4	Simplify Boolean function using the tabulation method $F(A, B, C, D) = \sum m(0, 1, 2, 3, 5, 6, 8, 9, 10, 12, 13, 14, 15)$		5				
450	4	Express the Boolean function $F(X, Y) = 1$ in canonical form.	A	1	$x'y' + x'y + xy' + xy$	xy	x + y	$X'Y' + XY + XY' + XY$
451	4	Given $F(A, B, C, D) = \sum m(2, 6, 8, 9, 10, 11) + \sum d(3, 7, 14, 15)$ is a Boolean function, where m represents min-terms and d represents don’t cares. The minimized logical function F is		1				
452	4	Express the Boolean function in sum of minterms and product of max terms. $F(W,X,Y,Z) = YZ + WXY + WXZ + WXZ$		3				
453	4	Minimize the logic function $F(A,B,C,D) = \pi M(1, 2, 3, 8, 9, 10, 11, 14).d(7, 15)$ Use Karnaugh map. Draw the logic circuit for simplified function using NOR gates only.		4				
454	4	Simplify Boolean function by using the tabulation method $F(A,B,C,D)=\sum m(2,3,6,7,8,9,10,11,12,13,14,15)$		5				
455	4	$F=\sum m(1,3,4,5,7,9,13)$ is a simplified version of the Boolean expression $D+A'.B$ only if there were a ‘don’t care’ entry. What is that don’t care term?	A	1	6,11,15	6,11,14	6,10,15	7,9,15
456	4	Which Boolean function does the following k-map represent? 	A	1	XNOR	AND	OR	NAND
457	4	Implement the following functions using the don’t-care conditions. $F = A'B'C' + AB'D + A'B'CD'$ $d = ABC + AB'D'$ Also implement the simplified expression using NAND gates only.		4				
458	4	Implement the following functions using the don’t-care conditions. $F = A'B'C' + AB'D + A'CD'$ $d = ABC + AB'D'$ Also implement the simplified expression using NOR gates only.		3				
459	4	Given the Boolean function F in three variables R, S and T as $F=R'ST'+RS'T+RST$ (a) Express F in the minimum product-of-sums form. (b) Assuming that both true and complement forms of the input variables are available, draw a circuit to implement F using the minimum number of 2 input NOR gates only		2				
460	4	Simplify the following function using Tabulation method: Show all steps along with PI chart and EPI terms. $F(W,X,Y,Z)=\sum m(0, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$		5				
461	4	Simplify the following function using Tabulation method: $F(W,X,Y,Z)=\sum m(0,1,2,5,6,7,8,9,10,14)$		5				
462	4	Express Function in Maxterms and Minterms: $F(x,y,z) = (xy + z)(y + xz)$		3				

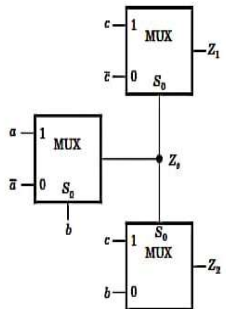
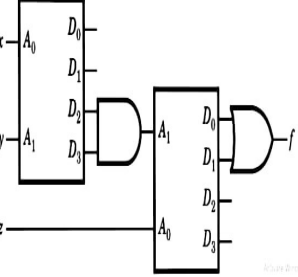
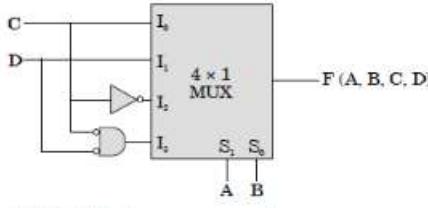
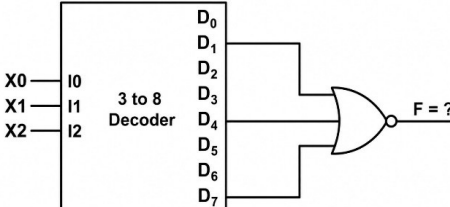
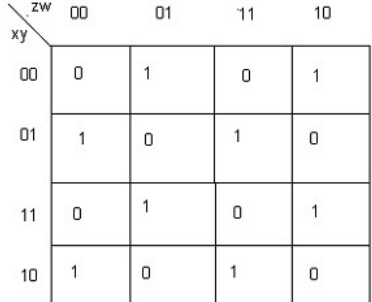
L J Institute of Engineering and Technology, Ahmedabad.								
Digital Electronics (DE) Practice Book (Sem-III 2026)								
Note: This Practice Book is only for reference purpose. LJU Test question paper may not be completely set from Practice Book.								
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463	4	In certain application, four inputs A, B, C, D (both true and complement forms available) are fed to logic circuit, producing an output F which operates a relay. The relay turns on when F(ABCD)=1 for the following states of the inputs (ABCD):'0000','0010' ,'0101','0110','1101' and '1110'. States '1000' and '1001' do not occur, and for the remaining states, the relay is off. Remaining states, the relay is off. Minimize F with the help of a Karnaugh map and realize it using a minimum number of 3- input NAND gates.		3				
464	5	A digital system consists of _____ types of circuits	A	1	2	3	4	5
465	5	Half-adders have a major limitation in that they cannot _____	C	1	Accept a carry bit from a present stage	Accept a carry bit from a next stage	Accept a carry bit from a previous stage	Accept a carry bit from the following stages
466	5	If A, B and C are the inputs of a full adder then the carry is given by _____	B	1	A AND B OR (A OR B) AND C	A AND B OR (A XOR B) AND C	(A AND B) OR (A AND B)C	A XOR B XOR (A XOR B) AND C
467	5	How many minimum AND, OR and EXOR gates are required for the basic configuration of full adder?	B	1	1, 2, 2	2, 1, 2	3, 1, 2	4, 0, 1
468	5	In a combinational circuit, the output at any time depends only on the _____ at that time.	C	1	Voltage	Intermediate values	Input values	Clock pulses
469	5	Procedure for the design of combinational circuits are: A. From the word description of the problem, identify the inputs and outputs and draw a block diagram. B. Draw the truth table such that it completely describes the operation of the circuit for different combinations of inputs. C. Simplify the switching expression(s) for the output(s). D. Implement the simplified expression using logic gates. E. Write down the switching expression(s) for the output(s).	C	1	B, C, D, E, A	A, D, E, B, C	A, B, E, C, D	B, A, E, C, D
470	5	Implement a combinational circuit of half adder using AOI gate.		3				
471	5	Implement a combinational circuit of half adder using NAND logic gates.		3				
472	5	Implement a combinational circuit of half adder using NOR logic gates.		3				
473	5	Implement a combinational circuit of full adder using only 2- input NAND logic gates.		4				
474	5	Implement a combinational circuit of full adder using only 2- input NOR logic gates.		4				
475	5	Implement a combinational circuit of full adder using AOI gates.		3				
476	5	Implement a combinational circuit of half subtractor using AOI gates.		3				
477	5	Implement a combinational circuit of half subtractor using NAND logic gates.		3				
478	5	Implement a combinational circuit of half subtractor using NOR logic gates.		3				
479	5	Implement a combinational circuit of full subtractor using only 2- input NAND logic gates.		4				
480	5	Implement a combinational circuit of full subtractor using only 2- input NOR logic gates.		4				
481	5	Implement a combinational circuit of full subtractor using AOI gates.		3				
482	5	Show how a full-adder can be converted to a full-subtractor with the addition of inverter circuit.		5				
483	5	With logic circuit describe the function of: Full adder,write the simplified Boolean functions for its outputs.		5				
484	5	Design the truth table of full adder and write the simplified Boolean function for its outputs.		4				
485	5	Design a full adder circuit using two half adders and gates.		4				
486	5	Differentiate between serial and parallel adders.		3				
487	5	In a half-subtractor circuit with X and Y as inputs, the Borrow (M) and Difference (N = X - Y) are given by	M=X'Y,N=X⊕Y	1	M=X⊕Y,N=XY	M=XY,N=X⊕Y	M=X'Y,N=X⊕Y	M=X'Y,N=(X⊕Y)'
488	5	A combinational logic circuit has memory characteristics that "remember" the inputs after they have been removed.	False	1				
489	5	To implement the full-adder sum functions, two exclusive-OR gates can be used.	True	1				
490	5	A half-adder does not have _____.	A	1	carry in	carry out	two inputs	all of the above
491	5	Which of the statements below best describes the given figure? 	A	1	Half-carry adder; Sum = 0, Carry = 1	Half-carry adder; Sum = 1, Carry = 0	Full-carry adder; Sum = 1, Carry = 0	Full-carry adder; Sum = 1, Carry = 1
492	5	A full-adder adds _____.	A	1	two single bits and one carry bit	two 2-bit binary numbers	two 4-bit binary numbers	two 2-bit numbers and one carry bit
493	5	The carry propagation delay in 4-bit full-adder circuits:	A	1	is cumulative for each stage and limits the speed at which arithmetic operations are performed	is normally not a consideration because the delays are usually in the nanosecond range	decreases in direct ratio to the total number of full-adder stages	increases in direct ratio to the total number of full-adder stages, but is not a factor in limiting the speed of arithmetic operations
494	5	Two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder. The carry input is 1. What are the values for the sum and carry output?	C	1	Σ4 Σ3 Σ2 Σ1=0111, C _{out} = 0	Σ4 Σ3 Σ2 Σ1 = 1111, C _{out} = 1	Σ4 Σ3 Σ2 Σ1= 1011, C _{out} = 1	Σ4 Σ3 Σ2 Σ1= 1100, C _{out} = 1
495	5	A full-adder has a C _{in} = 0. What are the sum and the carry (C _{out}) when A = 1 and B = 1?	B	1	Σ= 0, C _{out} = 0	Σ= 0, C _{out} = 1	Σ= 1, C _{out} = 0	Σ= 1, C _{out} = 1
496	5	An adder in which bits of the operand are added one after another _____	C	1	Half adder	Half subtractor	Serial Adder	parallel Adder
497	5	What is the major difference between half-adders and full-adders?	C	1	Full-adders are made up of two half-adders	Full adders can handle double-digit numbers	Full adders have a carry input capability	Half adders can handle only single-digit numbers
498	5	A combinational circuit calculates the arithmetic sum in a parallel way. What is the name of the adder?	B	1	Sequential Adder	Parallel Adder	Serial Adder	Both 1) & 2)
499	5	To implement a half adder and half subtractor ,the no. of basic(AND,OR,NOT) gates required is _____ and _____ respectively.	6,5	1				
500	5	A binary half-subtractor having two inputs A and B, the correct set of logical outputs D(=A minus B) and X(=borrow) are 1.The difference output D=A'B+AB' 2.The borrow output B=AB' Which of the above is/are correct?	A	1	1 Only	2 Only	none	Both 1 and 2
501	5	The number of full and half-adders required to add 16-bit numbers is	B	1	8half-adders, 8 full-adders	1 half-adder,15 full-adders	16 half-adders, 0 full-adders	4 half-adders, 12 full-adder

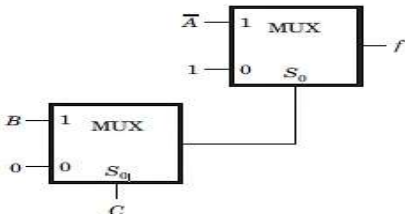
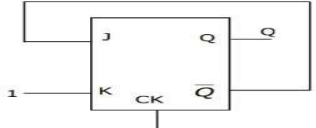
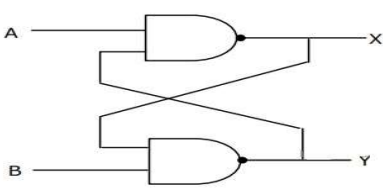
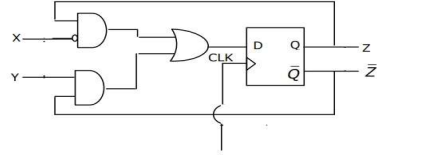
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502	5	How Serial Adder differ from Parallel Adders?	a serial adder processes bits one pair at a time sequentially over multiple clock cycles, while a parallel adder processes all bits simultaneously in a single clock cycle.	1				
503	5	The half-adder can be implemented from which of the following equation?	D	1	$S = xy' + x'y$ $C = xy$	$S = (x + y) (x' + y')$ $C = x y$	$S = (C + x'y')'$ $C = xy$	All of above
504	5	Full adder has ____	A	1	3 inputs	8 inputs	4 inputs	10 inputs
505	5	half adder is an example of-	A	1	Combinational circuit	Sequential circuit	Asynchronous circuit	None of these
506	5	How many full adders are needed to add two 4-bit numbers with a parallel adders?	B	1	2	4	16	8
507	5	The logic diagram shown in the figure performs the function of building block 	A	1	Half adder	Full adder	Half subtractor	Multiplier
508	5	Two binary digits are applied to input of two input Ex-or gate. The output of the logic can generate	C	1	Sum output	Difference output	Either sum or difference output	Carry output of a half adder
509	5	Two binary digits are applied to the input of two input AND gate. The output of the logic can generate	B	1	Borrow out of half subtractor	Carry out of half adder	Sum output of half adder	Difference output of half adder
510	5	Number of half and full adder required to construct a sixty four bit binary adder would be	A	1	One half adder and 63 full adders	64 full adders	64 half adders	one full adder and 63 half adder
511	5	A full subtractor can be constructed from two half subtractor and one _____	A	1	Two input OR gate	Two input AND gate	Two input Ex-or gate	Three input OR gate
512	5	A full adder can be constructed from two half adder and one _____	A	1	Two input OR gate	Two input AND gate	Two input Ex-or gate	Three input OR gate
513	5	What are the advantages of parallel adders over serial adders? Two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder. The carry input is 1. What are the values for the sum and carry output?		4				
514	5	Design Truth Table and give Boolean function for Full-Subtractor circuit and implement it using half subtractor and gates.		4				
515	5	Which of the following statements is/are not correct? 1) A half-adder is an arithmetic circuit that adds two binary digits. 2) A half-subtractor is an arithmetic circuit that subtracts two binary digits. 3) A full-subtractor is an arithmetic circuit that subtracts two binary digits along with borrow bit. 4) A full-adder is an arithmetic circuit that adds one binary digit and carry bit.	B	1	Only 1	Only 4	Only 3	Only 2
516	5	Which of the following is correct Boolean function of sum and carry of Full adder, the inputs are A, B and C (internal carry)?	C	1	$S = A \text{ xor } B \text{ xor } C$, $C_{out} = AB + BC' + AC$	$S = A \text{ xnor } B \text{ xor } C$, $C_{out} = AB + BC + AC$	$S = A \text{ xor } B \text{ xor } C$, $C_{out} = AB + (A \text{ xor } B).C$	$S = A \text{ xor } B \text{ xor } C$, $C_{out} = AB + (A \text{ xor } B').C$
517	5	Which of the following Boolean function of difference and borrow out (Bout) of full subtractor circuit, if the inputs are A, B and C (previous borrow)?	B	1	$D = A \text{ XOR } B \text{ XOR } C$, $B_{out} = A'B + AC + B'C'$	$D = A \text{ XOR } B \text{ XOR } C$, $B_{out} = A'B + A'C + BC$	$D = A \text{ XOR } B \text{ XOR } C$, $B_{out} = A'B + A'C + B'C$	$D = A \text{ XOR } B \text{ XOR } C$, $B_{out} = AB + AC + BC$
518	5	The logic diagram shown in the following figure performs the function of a very common arithmetic building block. Identify the logic function. 	A	1	Half subtractor	Full adder	Full subtractor	Half adder
519	5	In BCD addition, 0110 is required to be added to the sum for getting the correct result, if -	D	1	The sum of two BCD numbers is not a valid BCD number	The sum of two BCD number is greater than (1010)BCD only	a carry is produced	Both A and C
520	5	Design the digital circuit that adds (8) _{BCD} with (7) _{BCD} in parallel and produces a sum digit also in BCD format. Show the appropriate circuit diagram and steps that describe the BCD addition process.		4				
521	5	Design Binary Parallel Adder for addition of two binary numbers A= 1010 and B= 1110		2				
522	5	Design 4-Bit BCD adder which perform BCD addition of two decimal number 5 and 7.		3				
523	5	Design Logic diagram of Full adder and Full Subtractor with the help of truth table and logical equation.		4				
524	5	How many full adders are needed to add 4-bit numbers A and B with a Serial adder?	A	1	1	8	16	4
525	5	which of the following statement/s are true , I) In combinational circuits, memory are used for store the previous input so output of combinational circuits is depends on present as well as previous inputs. II) for n bit additions of binary number, number of full adder required for these is (n-1)n III) In full adder & full subtractor , there is similarity of sum bit (s) in case of full adder & borrow bit (bin) in case of full subtractor. IV) In output of BCD adder has range for sum output in 00000 to 10001.	D	1	i, ii, iii, iv	i, ii, iii	i, iii, iv	none of the above
526	5	Illustrate the adder circuit which adds two 4 bit numbers which are (3) _{BCD} & (9) _{BCD} through BCD addition & obtain the answer in BCD form.		5				
527	5	Design truth table and give Boolean function for Full Subtractor. Write the simplified Boolean functions for its outputs and implement it with NOR gates only.		3				

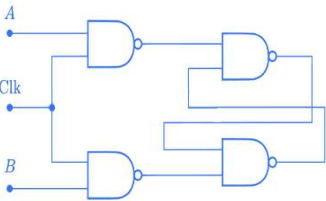
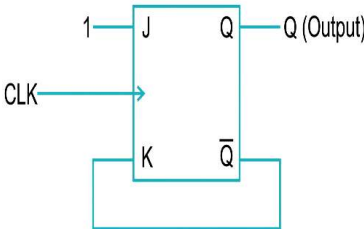
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
528	5	Illustrate the Addition of (15) ₁₀ & (7) ₁₀ By using binary adders of which requires an n-full adder for n-bit addition. (with logic diagram)		3				
529	5	Illustrate the combinational circuits that subtracts one bit from another one bit, when already there is a borrow from this column for subtraction in preceding column, and outputs the difference bit and borrow bit along with truth table, Boolean function & logic circuit diagram.		3				
530	5	Which statement/s is/are true? 1) A half adder is also known as XNOR gate because XNOR is applied to both inputs to produce the sum. 2) Half adder can add only two bits (A and B) and has nothing to do with the carry. 3) If the input to a half adder has a carry, then it will neglect it and adds only the A and B bits that means the binary addition process is not complete and that's way it is called a half adder. 4) A half adder is a two input and two output combinational circuit.	B	1	both 1 and 2	Three 2, 3 and 4	Three 1,3 and 4	All of the above
531	5	Design the digital circuit that adds (8) _{BCD} digit with (5) _{BCD} digit in parallel and produces a sum digit also in BCD.		4				
532	5	Design the digital circuit that adds (7) _{BCD} with (7) _{BCD} in parallel and produces a sum digit also in BCD format. Show the appropriate circuit diagram and steps that describe the BCD addition process		4				
533	5	Design a 4-bit parallel adder to find the sum and output carry for the addition of the following two 4-bit numbers which are A0A1A2A3 = 1011 and B0B1B2B3 = 1111 and the input carry (C0) is 0. List bitwise sum and output carry for the 4-bit parallel adder.		2				
534	5	Design the digital circuit that adds (7) _{BCD} with (9) _{BCD} in parallel and produces a sum digit also in BCD format. Show the appropriate circuit diagram and steps that describe the BCD addition process.		4				
535	5	Design a combinational circuit that generates BCD sum of (3) _{BCD} and (9) _{BCD} number with necessary equations and truth table.		3				
536	5	Design a combinational circuit that add (9) _{BCD} and (8) _{BCD} number with necessary equations.		4				
537	5	Prove that a full adder can be obtained by two half adder and an OR gate.		3				
538	6	_____ is the example of multiplexer.	D	1	2 x 1	1 x 2	4 x 1	both (a) & (c)
539	6	_____ is used for connecting two or more sources to a single destination among computer units.	B	1	De-multiplexer	Multiplexer	encoder	decoder
540	6	A multiplexer is also called as _____	C	1	data collector	data distributor	data selector	data controller
541	6	2 to 1 line multiplexer can be realized using _____	D	1	1 AND, 2 OR & 1 NOT gates	2 AND, 2 OR & 1 NOT gates	2 AND, 1 OR & 2 NOT gates	2 AND, 1 OR & 1 NOT gates
542	6	_____ is the major functioning responsibility of the multiplexing combinational circuit?	C	1	Decoding the binary information	Generation of all minterms in an output function with OR-gate	Generation of selected path between multiple sources and a single destination	Encoding of binary information
543	6	_____ logic inputs should be given to the input lines I ₀ , I ₁ , I ₂ and I ₃ , if the MUX is to behave as two input XNOR gate? 	B	1	110	1001	1010	1111
544	6	_____ is the function of an enable input on a multiplexer chip?	C	1	To apply Vcc	To connect ground	To active the entire chip	To active one half of the chip
545	6	A 4 × 1 MUX is used to implement a 3 - input boolean function as shown in figure. The boolean function F(A,B,C) implement is _____ 	A	1	$F(A,B,C) = \Sigma(1,2,4,6)$	$F(A,B,C) = \Sigma(1,2,6)$	$F(A,B,C) = \Sigma(2,4,5,6)$	$F(A,B,C) = \Sigma(1,5,6)$
546	6	In a multiplexer, the selection of a particular input line is controlled by _____	B	1	Data controller	Selection lines	Logic gates	Both data controller and selected lines
547	6	_____ select lines would be required for an 8-line-to-1-line multiplexer.	D	1	2	4	8	3
548	6	_____ NOT gates are required for the construction of a 4-to-1 multiplexer.	A	1	2	5	3	4
549	6	The output of the 4 to 1 MUX shown in figure is: 	B	1	$x' + y'$	$x + y$	$xy + x'$	$(xy)' + x$

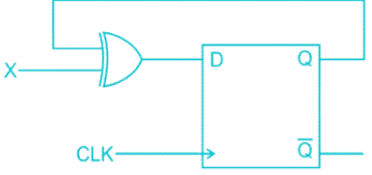
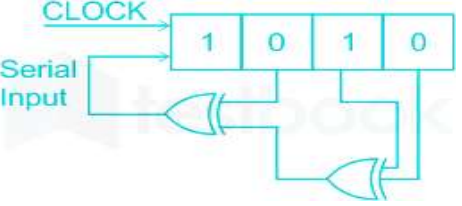
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
550	6	_____ is the output of design. 	A	1	$w'xy'z' + wx'y'z' + xy + yz$	$wx'yz + w'xyz + x'y' + y'z'$	$w'xy'z' + w'x'y'z + yz' + zx$	$w'xyz + wxyz' + gz + z'x'$
551	6	Design and elaborate 2 x 1 multiplexer.		4				
552	6	Implement and elaborate 4 x 1 multiplexer.		4				
553	6	Construct and elaborate 8 x 1 multiplexer.		4				
554	6	Realize the following function of three variables with 8:1 MUX. $F(A, B, C) = \Sigma(0, 1, 3, 4, 7)$		4				
555	6	Obtain an 8 to 1 multiplexer with a dual 4 to 1 line multiplexers having separate enable inputs but common selection lines. Use a block diagram construction.		5				
556	6	Implement $F(P, Q, R) = PR' + P'QR' + Q'R + P'$ by using 3x8 line decoder.		2				
557	6	Illustrate the comparison circuit of two numbers A & B having the bit value is 3 bits as A2A1A0 and B2B1B0, using suitable comparator with block diagram and logic diagram.		3				
558	6	Implement $F(W, X, Y, Z) = \Pi M(1, 4, 7, 10, 12)$ using 8x1 MUX having Y as input to MUX and W, X and Z as selection lines of MUX.		3				
559	6	Implement the following function with a multiplexer: $F(A, B, C, D) = \Sigma(0, 1, 3, 4, 8, 9, 15)$		4				
560	6	A 4 × 1 MUX is shown in the figure below, the output Z is _____ (Where S1 is MSB and S2 is LSB) 	D	1	$F(A, B, C) = A \oplus C$	$F(A, B, C) = A \odot C$	$F(A, B, C) = (A \odot C)'$	Both A and C
561	6	A, B and Cin are the three inputs of a full adder circuit and D0, D1, ..., D7 are the inputs of 8×1 multiplexer. S2 (MSB), S1 and S0 (LSB) are the selection lines of the multiplexer. To implement the expression of sum of full adder circuit using this multiplexer, the connections of the input ports and selection lines are	A	1	$D0=D3=D5=D6=0, D1=D2=D4=D7=1, S2=A, S1=B \text{ and } S0=Cin$	$D0=D3=D5=D6=1, D1=D2=D4=D7=0, S2=Cin, S1=B \text{ and } S0=A$	$D0=D2=D3=D6=0, D1=D4=D5=D7=1, S2=A, S1=B \text{ and } S0=Cin$	$D0=D1=D5=D7=1, D2=D3=D4=D6=0, S2=Cin, S1=B \text{ and } S0=A$
562	6	There are eight news channels are connected at 8:1 MUX with the selection line of S2, S1, S0: if Ch-1 as “AAJ TAK”, Ch-2 as “NDTV”, Ch-3 as “24x7”, Ch-4 as “ZEE NEWS”, Ch-5 as “INDIA TV”, Ch-6 as “BBC NEWS”, Ch-7 as “TV18”, Ch-8 as “DD NEWS” and Selection lines are S2, S1, S0 as 101 so which channel would be selected:	C	1	AAJ TAK	NDTV	BBC NEWS	TV18
563	6	_____ is the example of Demultiplexer.	B	1	2 x 1	1 x 2	4 x 1	both (a) & (c)
564	6	How many selection lines required in a single input n- output demultiplexer ?	D	1	2	n	2^n	log n(base 2)
565	6	The word demultiplex means _____	D	1	One into many	Many into one	Distributor	One into many as well as Distributor
566	6	In 1-to-4 demultiplexer, _____ select lines are required	A	1	2	3	4	5
567	6	Most demultiplexers facilitate _____ type of conversion	B	1	Decimal-to-hexadecimal	Single input, multiple outputs	AC to DC	Odd parity to even parity
568	6	Design and elaborate 1 x 2 demultiplexer.		4				
569	6	Implement and elaborate 1 x 4 demultiplexer.		4				
570	6	Construct and elaborate 1 x 8 demultiplexer.		4				
571	6	_____ converts binary coded information to unique outputs such as decimal, octal digitals etc.	A	1	decoder	demultiplexing	multiplexing	encoder
572	6	In a decoder, if the input lines are 4 then _____ number of maximum output lines .	B	1	2	16	4	8
573	6	_____ can be represented for decoder.	B	1	Sequential circuit	Combinational circuit	Logical circuit	None of the mentioned
574	6	BCD to seven segment conversion is a _____	A	1	Decoding process	Encoding process	Comparing process	None of these
575	6	With which decoder it is possible to obtain many code conversions?	D	1	2 line to 4 line	3 to 8 line	not possible with any decoder	4 to 16 line
576	6	A device which converts BCD to seven segment is called as _____.	C	1	multiplexer	encoder	decoder	inverter
577	6	For design 3 to 8 decoder _____ 2 to 4 decoder is required.	D	1	3	5	4	2
578	6	Decoder is constructed from _____.	C	1	Inverters	AND gates	Inverters and AND gates	None of the mentioned
579	6	Design and elaborate 3 to 8 line decoder.		5				
580	6	Implement half adder using 2 × 4 line decoder.		2				
581	6	Implement full adder circuit using 3 × 8 line decoder		4				
582	6	Design and elaborate 2 to 4 line decoder.		5				

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
583	6	Consider the logic circuit given below. The minimized expression for F is ____ 	A	1	C'	Io	C	Io'
584	6	Implement $A + B'C'$ by using 3×8 line decoder.		3				
585	6	Construct binary to octal decoder and justify with block diagram.		4				
586	6	For 8-bit input encoder _____ combinations are possible	B	1	8	2^8	4	2^4
587	6	Design Octal to Binary encoder.		4				
588	6	Design 4×2 encoder logic diagram with truth table.		5				
589	6	Implement 8×3 encoder logic circuit along with truth table.		5				
590	6	_____ is the 4 bit gray code of binary number 1010 .	B	1	1110	1111	1101	1011
591	6	_____ is the binary number of 1110 gray code.	A	1	1011	1101	1010	110
592	6	Design and implement a 4 bit binary to gray code converter		5				
593	6	Design and implement a 4 bit gray to binary code converter		5				
594	6	Design and implement a BCD to seven segment code converter		4				
595	6	If A & B is 1 and 0 respectively then ____ will be output with reference to 1 bit comparator.	B	1	$A = B$	$A > B$	$A < B$	$A'B$
596	6	_____ gate is basic comparator.	C	1	NOR	NAND	XNOR	NOT
597	6	Two inputs are $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ gives 4-bit equality condition then _____ will be implementation.	B	1	$A \cdot B$	$A \odot B$	$A + B$	$A > B$
598	6	When in comparator A & B is coincide then _____ is the output.	C	1	$A > B$	$A < B$	$A = B$	$A.B$
599	6	Implement 1- bit magnitude comparator with logic diagram.		4				
600	6	Design 2- bit comparator with logic diagram.		5				
601	6	_____ number of x-or gates used in 3 bit even parity generator.	D	1	3	4	5	2
602	6	Design an SOP circuit that will generate an odd parity bit for 3-bit input.		5				
603	6	Implement logic circuit for the 3-bit even parity generator along with k-map.		5				
604	6	Design for $P = \sum m(0, 3, 5, 6)$ parity generator with logic diagram.		5				
605	6	_____ variables required for 4 bit even parity checker.	D	1	1	2	3	4
606	6	_____ times 1's possible in 4-bit even parity checker.	A	1	8	7	6	9
607	6	For error detection and correction codes , _____ functions are very useful in the system.	C	1	NAND	NOR	EX-OR	OR
608	6	_____ number of x-or gates required in 4 bit even parity checker.	B	1	2	3	4	5
609	6	Design k- map for 4- bit odd parity checker.		4				
610	6	Design SOP expression of $P = \sum m(0, 3, 5, 6, 9, 10, 12, 15)$ parity checker with k-map and logic diagram.		6				
611	6	Design a combinational circuit and evaluate expression for binary to gray code converter for given binary input B3B2B1B0 and gray output G3G2G1G0.		4				
612	6	Design full subtractor logic circuit using 3x8 decoder and OR gates. Use X, Y and Z as input variables and D & B as output variables		3				
613	6	Which logic gate is generated for multiplexer inputs B' (MSB) and 1 (LSB) having selection line A?	A	1	NAND	OR	NOR	XNOR
614	6	How many 2x1 multiplexers are required for implementing 16x1 multiplexer?	C	1	12	11	15	16
615	6	If the number of n generated output lines is equal to 2^m in demultiplexer then it requires _____ selection lines.	C	1	2^m	2^n	m	n
616	6	In the circuit shown, W and Y are MSB's of select inputs. The output function F is given by - 	D	1	$(W \text{ XNOR } X)Y'$	$(W \text{ XOR } X)Y$	$(W \text{ XNOR } X)Y$	$(W \text{ XOR } X)Y'$
617	6	Which of the above statement/s is/are correct? 1. A 8-to-3 line decoder is also called as binary to octal decoder. 2. A DEMUX circuit is also known as data selector circuit. 3. A 8-to-1 MUX routes the desired data input to the output is controlled by 2 select lines. 4. For an n:1 multiplexer the number of select lines are given by $\log_2 n$	D	1	Both 1 and 2	Both 1 and 4	only 3	None of the given
618	6	Design two input NOR gate and XNOR gate using 2-to-1 line Multiplexer.		2				
619	6	Implement the function $F(A,B,C,D) = A'B + CD' + AC'$ using MUX with 3 select lines. Use B, C, and D as select lines.		3				
620	6	The Combinational Logic Circuit with the function of P (A, B, C) = $\sum m(1, 2, 4, 7)$ is _____ parity generator with expression _____	A	1	Even Parity, A XOR B XOR C	Odd Parity, A XOR B XOR C	Even Parity, A XNOR B XOR C	Odd Parity, A XOR B XNOR C
621	6	Design a Combination Logic Circuit which accepts 3-Bit gray number and provide its binary equivalent as output.		3				
622	6	Construct Binary to Octal Decoder along with truth-table and equation. Also Implement Full adder using decoder.		3				
623	6	Design a combinational logic circuit that compare single bit binary numbers A and B to check if they are equal, greater or less.		3				

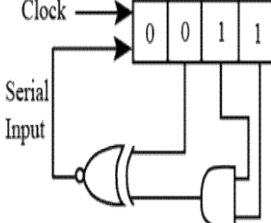
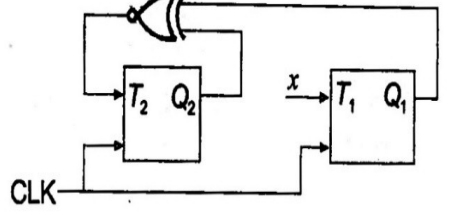
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
624	6	A MUX network is shown in fig. What is the value of output Z_1 = ? 		1				
625	6	A logic circuit of 2 x 4 decoder shown in figure , output of decoder are as follow : $D_0 = A'0 A'1$, $D_1 = A'0 A_1$, $D_2 = A_0 A'1$, $D_3 = A_0 A_1$. Then value of $f(x,y,z)$ =..... 	C	1	0	z	z'	1
626	6	The output Y of a 2-bit comparator is logic 1 whenever the 2- bit input A is greater than the 2-bit input B. The number of combination for which the output is logic is 1 is __	B	1	4	6	8	10
627	6	During the design of BCD to 7 segment decoder (or converter) , what value is initialized to display 9 ? (options are in decimal equivalent of outputs of decoder)	B	1	126	123	127	115
628	6	Design NAND logic gate & NOR logic gate using 2x1 multiplexer.		3				
629	6	The logic function $F(A,B,C,D) = AC + ABD + ACD$ is to be realised using 8x1 mux using the variable B,C,D as control input.		3				
630	6	Illustrate the comparison circuit of two numbers A & B having the bit value is 1 nibble to each numbers, using suitable comparator with block diagram and logic diagram.		5				
631	6	The Boolean function realized by the logic circuit shown is 	A	1	$F = \sum m(2,3,5,7,8,9,12)$	$F = \sum m(2,3,4,5,7,8,9,12)$	$F = \sum m(2,3,5,8,9,12)$	$F = \sum m(2,3,5,7,8,9,11)$
632	6	Find F for given circuit. 	D	1	$F = \sum m(1,4,7)$	$F = \sum m(1,2,4,7)$	$F = \sum m(1,2,4,5,6)$	$F = \sum m(0,2,3,5,6)$
633	6	Identify that this K-Map belongs to which digital circuit? 	C	1	Odd parity generator	Odd parity checker	Even parity checker	Even parity generator
634	6	Design a combinational circuit with three inputs x,y, and z and three outputs A,B, and C. when the binary input is 0,1,2 or 3, the binary output is one greater than the input. When the binary input is 4,5,6 or 7, the binary output is one less than the input.		4				
635	6	Implement the function $F(W, X, Y, Z) = WX + WYZ + XYZ$ using MUX with 3 select lines. Use W, Y, and Z as select lines and X as a Data input.		4				

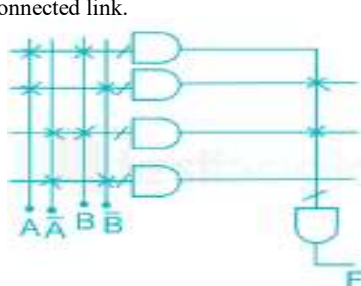
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
636	6	Design a combinational logic circuit which converts 4-bit Gray input (G3G2G1G0) to binary output (B3B2B1B0) with necessary equations.		4				
637	6	Design a full subtractor circuit using 3*8 decoder with its diagram.		3				
638	6	A magnitude comparator is a digital comparator which has ____ output terminals.	C	1	1	2	3	4
639	6	A logic circuit takes 4-BCD inputs {A, B, C and D) to give an output F. Output F is '1' if the input is an invalid BCD-code. The minimized Boolean equation is ____	B	1	AC+BD	AB+AC	B'C+AD'	A'B+A'C
640	6	The network shown in given figure is ____ 	B	1	OR gate	NAND gate	XOR gate	NOR gate
641	6	A 16-to-1 multiplexer is to be implemented using only 4-to-1 multiplexers. What is the minimum number of 4-to-1 multiplexers required?	D	1	2	3	4	5
642	6	Design a full subtractor circuit using 4*1 multiplexer. Take A, B and C as inputs of full subtractor where A and C as selection lines.		3				
643	6	Design a combinational logic circuit that compare 3-bit binary numbers X and Y to check if they are equal, greater or less.		5				
644	7	Which of the following flip-flops is free from the race around problem?	C	1	T flip-flop	SR flip-flop	Master-Slave Flip-flop	D flip-flop
645	7	The characteristic equation of S-R latch is _____	A	1	$Q(n+1) = S + Q(n)R'$	$Q(n+1) = SR + Q(n)R$	$Q(n+1) = S'R + Q(n)R$	$Q(n+1) = S'R + Q'(n)R$
646	7	D flip flop can be made from a J-K flip flop by making	D	1	J=K	J=K=1	J=0,K=1	J=K'
647	7	In a J-K flip flop, when $J_n = 0$ and $K_n = 1$, the output Q_{n+1} will have a value of:	B	1	1	0	Q_n	Q_{n+1}
648	7	In JK flip flop, which combination will toggle the output?	B	1	J=0,K=0	J=1,K=1	J=1,K=0	J=0,K=1
649	7	Master slave flip flop is also referred to as?	B	1	Level triggered flip flop	Pulse triggered flip flop	Edge triggered flip flop	Edge-Level triggered flip flop
650	7	In a positive edge triggered JK flip flop, a low J and low K produces?	D	1	High state	Low state	Toggle state	No Change State
651	7	How is a J-K flip-flop made to toggle?	D	1	J = 0, K = 0	J = 1, K = 0	J = 0, K = 1	J = 1, K = 1
652	7	D flip-flop is a circuit having _____ along with one inverter.	C	1	2 NAND gates	3 NAND gates	4 NAND gates	5 NAND gates
653	7	When is a flip-flop said to be transparent?	B	1	When the Q output is opposite the input	When the Q output follows the input	When you can see through the IC packaging	When the Q output is complementary of the input
654	7	On a positive edge-triggered S-R flip-flop, the outputs reflect the input condition when _____	C	1	The clock pulse is LOW	The clock pulse is HIGH	The clock pulse transitions from LOW to HIGH	The clock pulse transitions from HIGH to LOW
655	7	The output Q_n of J-K flip flop is 1. It changes to 0 when a clock pulse is applied. The inputs J_n and K_n are respectively	C	1	0 and X	1 and X	X and 1	X and 0
656	7	In a J-K flip flop, when J = 1 and K = 1 then it will be considered as	D	1	set conditions	reset conditions	no change	toggle condition
657	7	When _____ the outputs either don't change or they are invalid in SR latch.	C	1	S=R=0	S=R=1	Both A & B	S=0, R=1
658	7	Which of the following logic circuits do not have no-change condition?	D	1	JK-FF	SR-FF	T-FF	D-FF
659	7	In J-K-M-S flip flop comprises _____ followed by _____ configuration?	D	1	S-R flip-flop, S-R flip flop	S-R flip-flop, J-K flip flop	J-K flip-flop, J-K flip flop	J-K flip-flop, S-R flip flop
660	7	Which flip flop can be obtained from an S-R flip flop by just putting one Inverter between S and R?	B	1	T	D	J-K	latch
661	7	If a JK FF toggles more than once during one clock cycle it is called	A	1	racing	pinging	spiking	bouncing
662	7	In a D flip flop the output state Q is related with D input in what way?	A	1	Q is same as D	Q is complement of D	Q is independent of D	Q is dependent of D
663	7	In a J_K flip-flop, we have J= Q' and K=1 (see figure). Assuming the flip-flop was initially cleared and then clocked for 6 pulses, the sequence at the Q output will be 	D	1	010000	011001	010010	010101
664	7	In figure, A = 1 and B = 1. The input B is now replaced by a sequence 101010 ____, the outputs x and y will be 	A	1	fixed at 0 and 1, respectively	x = 1010.... While y = 0101....	x = 1010.... While y = 10101....	fixed at 1 and 0, respectively
665	7	A sequential circuit using D flip-flop and logic gates is shown in the figure, where X and Y are the inputs and Z is the output. The circuit is 	D	1	S-R Flip flop with inputs X = R and Y = S	S-R Flip flop with inputs X = S and Y = R	J-K Flip flop with inputs X = J and Y = K	J-K Flip flop with inputs X = K and Y = J

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
666	7	In a master-slave flip-flop, when is the master enabled?	B	1	when the gate is LOW	when the gate is HIGH	both of the above	neither of the above
667	7	Consider the given circuit. In this circuit, the race around 	A	1	Does not occur	occurs when CLK = 0	occurs when CLK = 1 and A=B =1	occurs when CLK = 1 and A=B =0
668	7	Master slave configuration is used in flip flop to	C	1	increase its clocking rate	reduces power dissipation	eliminates race around condition	improve its reliability
669	7	Initial state: Q = 0 The output sequence Q of the circuit shown below is– 	D	1	0000	0001110...	11001100	11111
670	7	In T flip-flop the output frequency is—	C	1	Same as the input frequency	Double of its input frequency	One-half its input frequency	None of these
671	7	A 2MHz clock is applied to J=K=1. What is the frequency of flip flop O/P signal?	B	1	2MHz	1MHz	500 Hz	0
672	7	The output Q _{n+1} of a J-K flip-flop for the input J =1, K=1 is	D	1	0	1	Q _n	Q _n '
673	7	The characteristic equation of J-K flip-flop is _____	D	1	$Q(n+1)=JQ(n)+K'Q(n)$	$Q(n+1)=JQ'(n)+KQ(n)$	$Q(n+1)=JQ'(n)+KQ(n)$	$Q(n+1)=JQ'(n)+K'Q(n)$
674	7	Find Excitation table of JK,D,T,SR flop flop from their truth table.		4				
675	7	Convert JK flip flop to T flip flop.		4				
676	7	Design master slave JK flipflop.		4				
677	7	Convert T Flipflop in to D flipflop.		4				
678	7	Discuss working of clocked delay type flip-flop with characteristic table and logic diagram.		4				
679	7	Distinguish between combinational and sequential logic circuits.		4				
680	7	Distinguish between latch and flipflops.		2				
681	7	Convert SR flip-flop into JK flip-flop.		4				
682	7	Draw the circuit diagrams and Truth table of all the Flip flops (SR, D, T and JK).		4				
683	7	Implement D flip flop using JK flip flop.		4				
684	7	Convert JK flip flop to SR flip flop.		4				
685	7	Convert D flip flop into SR flip flop.		4				
686	7	Implement T flip flop using D flip flop.		4				
687	7	What is race-around condition in JK flip-flop?		2				
688	7	Design edge triggered flip flop in detail.		4				
689	7	Design S R flip flop usingNAND and NOR gate		4				
690	7	Convert D flip flop into JK flip flop.		4				
691	7	Which flipflop is generated by shorting the inputs of JK flipflop? Evaluate the characteristic equation and give excitation table of generated flipflop.		4				
692	7	Which of the following is/are correct for a D-type flip flop?	C	1	The output toggles if one of the input is held HIGH	The one of the output can be invalid condition	The Q output is either SET or RESET as soon as the D input goes HIGH or LOW	Both inputs are always same with each other
693	7	The output Q _n of an SR flip-flop is 1. It changes to 0 when a clock pulse is applied. The inputs S and R respectively –	B	1	X and 1	0 and 1	X and 0	1 and X
694	7	The invalid state of NAND based SR latch occurs when –	B	1	S = 1, R = 1	S = 0, R = 0	S = 1, R = 0	S = 0, R = 1
695	7	Consider the following statements: 1. Race-around condition occurs in a JK flip-flop when the inputs are 1, 1 and CLK = 1. 2. A flip-flop is used to store one bit of information. 3. In active-HIGH SR latch, Set = 1, Reset = 0 indicates RESET state. 4. Master-slave configuration is used in a flip-flop to store 2-bits of information. Which of the above statement/s is/are incorrect?	B	1	Both 1 and 2	Both 3 and 4	Both 2 and 3	all are incorrect
696	7	In a JK flip-flop, output Q _n = 1 and it does not change when a clock pulse is applied. What is the possible combination of inputs (J _n , K _n) in this condition?	A	1	(X,0)	(X,1)	(1,0)	(1,X)
697	7	Consider two flip flops X and Y. Flip flop X has single input in which output (Q _{n+1}) follows the input. Flipflop Y has two inputs and output, becomes invalid when both inputs are HIGH. Convert Flip flop X into Y.		4				
698	7	Consider two flip flops A and B. Flip flop A has a single input and positive edge triggered clock pulse, and has a property that output of the next state (Q _{n+1}) is equal to the input. Flip-flop B contains 2 input terminals and positive edge triggered clock pulse. Flip-flop B toggles more than once during one clock cycle. Convert the flip-flop A into B and also show the required steps of conversion.		4				
699	7	In a positive edge triggered T Flip Flop, high T produces?	A	1	TOGGLE STATE	NO CHANGE STATE	INVALID STATE	UNSTABLE STATE
700	7	Which of the following statement/statements is/are true for Master-Slave Flip-Flop? 1) It is a pulse-triggered Flip-Flop 2) This can be used to eliminate the race-around condition. 3) Master is enabled when the clock is at a low level	A	1	ONLY 1 AND 2	ONLY 1, 2 AND 3	ONLY 3	ONLY 2
701	7	Implement SR Flip-Flop using Gated Transparent Latch		3				
702	7	Design two input SET-RESET Flip-Flop which eliminates invalid state and formulate its characteristic equation with the help of truth table.		3				

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
703	7	Choose the correct statement/s from following . 1) In SR-latch using NAND gates, set & reset input are normally in low state & one of them will be pulsed high whenever we want to change latch output. 2) In SR-lath using NOR gates, set & reset input are normally in high state & one of them will be pulsed low whenever we want to change latch output. 3) Race around condition can be avoided by using level triggered JK flipflop.	D	1	1,2,3	2 & 3	1 & 3	none of above
704	7	Illustrate the SR-latch using active high input with its logic diagram and truth table & state that how flip flop is differed from the latch.		3				
705	7	Convert JK flip flop to SR flip flop with the help of characteristics table and excitation table of flipflops & draw the equivalent of logic diagram of converted flip flop.		5				
706	7	In a JK-FF, When $J_n=0$ and $K_n=1$, the output Q_{n+1} will have a value of ____	B	1	1	0	Q_n	Q_{n+1}
707	7	How JK-FF can be converted into SR-FF?		4				
708	7	Which flip flop is generated by sorting the inputs of JK flipflop? Evaluate the characteristics equation and give excitation table of generated flipflop.		3				
709	7	Design a pulse triggered flipflop which can eliminates race around condition.		2				
710	7	Discuss how SR flipflop can be made using JK Flipflop.		3				
711	7	____ are the applications of flipflops.	D	1	Storage Devices	Data Transfers	Register	All of above
712	7	The equivalent flip – flop by the circuit shown below is 	C	1	JK	SR	T	Master Slave
713	7	Latches constructed with NOR and NAND gates tend to remain latched condition due to which configuration?	B	1	Gate Impedance	Cross Coupling	Synchronous operation	Asynchronous operation
714	7	An JK flipflop can be converted to T flipflop by connecting ____	D	1	J to Q	K to Q	K to Q'	J to K
715	7	The present output Q_n of an edge triggered JK flip-flop is logic 0. If $J=1$, then Q_{n+1} is ____.	C	1	cannot be determined	will be logic 0	will be logic 1	will race around
716	7	What is the characteristic of a master-slave flip-flop?	C	1	Change in the output occurs when the state of the master is affected.	Both the master and the slave states are affected at the same time.	Change in the output occurs when the state of the slave is affected.	Change in the input immediately reflected in the output
717	7	Which flip flop can be obtained from an S-R flip flop by just putting one Inverter between S and R? Discuss characteristic equation and excitation table of obtained flipflop.		3				
718	8	Design 4-bit Universal Shift Register.		5				
719	8	An 8 bit serial in/ serial out shift register is used with a clock frequency of 2MHz to achieve a time of delay of	C	1	16 μ s	8 μ s	4 μ s	2 μ s
720	8	With a 200 khz clock frequency, eight bits can be serially entered into a shift register in	B	1	4 μ s	40 μ s	400 μ s	40ms
721	8	A serial in/parallel out initially contains all 1s. The data nibble 0111 is waiting to enter, after four clock pulses, the register contains	C	1	0000	1111	0111	1000
722	8	Assume that a 4 bit serial in/ serial out shift register is initially clear. we wish to store the nibble 1100. what will be the 4 bit pattern after the second clock pulse ?	C	1	1100	0011	0000	1111
723	8	SIPO is a abbreviation of	A	1	serial in parallel out	parallel in serial out	serial in serial out	serial in peripheral out
724	8	A function of register is	A	1	Store data	multiplex a data	demultiplex a data	decode a data
725	8	Design 4-bit bidirectional shift register with Parallel load.		4				
726	8	A bidirectional 4 bit nibble is storing the nibble 1110. its input is low. the nibble 0111 is waiting to be entered on the serial data input line. after two clock pulse the shift register is storing	D	1	1110	0111	1000	1001
727	8	The group of bits 10110111 is serially shifted(right most bit first) into an 8bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains	D	1	10111000	10110111	11110000	11111100
728	8	Show the working of Universal Shift Register using symbol, block diagram and truth table		5				
729	8	Design a 4-bit Universal register using combinational circuit which satisfies following conditions: S1 S0="10" for No Change, S1 S0= "00" for Shift Left, S1 S0="01" for Parallel load and S1 S0="11" for Shift Right.		5				
730	8	Design a circuit for 4-bits parallel register with load with D Flip-Flops. Load input decides whether to load new input or to apply no change conditions.		5				
731	8	True or False. Data can be changed from special code to temporal code by using Counter.	False (Using Shift register)	1	TRUE	FALSE		
732	8	True or False. A universal shift register has both serial and parallel input and output capacity.	True	1	TRUE	FALSE		
733	8	The shift register is initially loaded with bit pattern 1010. Shift register is clocked in which count gets shifted by one position to right with each shift. The bit at serial input is pushed to left most position(MSB). After how many clock pulse will the content of shift register will be 1010? 	B	1	3	7	11	15
734	8	Which of the following is not the characterstic of shift register?	A	1	Serial in/parallel in	Serial out/parallel in	Serial in/parallel out	Parallel in/parallel out
735	8	True or False. A counter has a specified sequence of states, but a shift register does not.	True	1	TRUE	FALSE		
736	8	Explain the working of 4 bit asynchronous counter.		4				

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Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
737	8	Design 4-bit binary ripple counter		4				
738	8	Enlist the classification of counters and Design asynchronous 4-bit binary ripple counter.		4				
739	8	Design 4-bit up-down binary synchronous counter.		4				
740	8	Design 3 bit binary counter with necessary diagrams.		4				
741	8	Design BCD synchronous Counter.		4				
742	8	Design a synchronous Counter that goes 0,2,3,7,0,2,3,...		5				
743	8	Demonstrate about a synchronous counter using 3 bits.		5				
744	8	Design 3-bit synchronous up counter using D flip flop.		4				
745	8	Justify:- A counter works as frequency divider with suitable example.		4				
746	8	Design a Mod-13 Asynchronous counter using Negative edge triggered T Flip flop with necessary waveforms		6				
747	8	Design a synchronous counter which goes through sequence 0,7,2,5,1,6,4 and repeat using positive edge triggered SR flip-flops.		6				
748	8	Design a synchronous BCD counter with D flip flops.		4				
749	8	Design a mod-12 Synchronous up counter using D-flipflop.		4				
750	8	Design and implement a Modulo – 6 Asynchronous counter using JK flipflop.		4				
751	8	Design and implement a Modulo – 6 Synchronous counter using JK flipflop.		4				
752	8	Enlist the significant difference between synchronous and asynchronous counters.		4				
753	8	Design BCD ripple counter with its operation and develop its logic diagram.		4				
754	8	With logic diagram explain the operation of 4 bit binary ripple counter. Explain the count sequence. How up counter can be converted into down counter?		5				
755	8	Design and explain 4 – bit Ripple UP/ DOWN counter using positive edge triggered Flip flop.		5				
756	8	Design a counter to generate the repetitive sequence 0,1,2,4,3,6.		5				
757	8	Design Modulo-8 counter using D flip flop.		5				
758	8	Design a counter that counts the sequence as 0, 1, 2, 4, 5, 6 and rolls over to 0 again. Use +ve edge triggered D-FF or JK-FF.		5				
759	8	Design and explain BCD Counter.		5				
760	8	Design 4 bit ripple counter using timing diagrams.		5				
761	8	Design 3-bit ripple up-counter using negative edge triggered T-FF or JK- FF . Also draw the waveforms..		5				
762	8	Design a counter to generate the repetitive sequence 0, 3, 5, 7, 4 using D FF.		5				
763	8	Design sequential counter for sequence 0,2,4,3,7,6,1 using JK flip-flops.		5				
764	8	Design a 3-bit synchronous up counter using K-maps and positive edge-triggered JK FFs.		5				
765	8	Design and explain 4-bit Ripple UP/DOWN Counter using negative edge triggered Flipflop.		5				
766	8	Design and implement a Modulo-4 Asynchronous counter using T Flip flop.		5				
767	8	Design and implement a Modulo-6 Asynchronous counter using T Flip flop.		5				
768	8	Design 4-bit ripple counter using negative edge triggered JK flip flop.		5				
769	8	What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of ‘n’ number of flip-flops?	C	1	0 to 2 ⁿ	0 to 2 ⁿ +1	0 to 2 ⁿ - 1	0 to 2 ^{n+1/2} - 1
770	8	Design a counter for following binary sequence 0-1-3-4-6-0		5				
771	8	In a 4 bit johnson counter sequence, there are a total of how many states or bit patterns	D	1	1	3	4	8
772	8	If a 10 bit ring counter has an initial state 1101000000, what is the state after the second clock pulse?	B	1	1101000000	11010000	1100000000	1100000011
773	8	On the fifth clock pulse, a 4-bit Johnson counter sequence is Q0 = 1, Q1 = 1, Q2 = 0, and Q3 = 0. On the eleventh clock pulse, the sequence is _____.	A	1	0000	0001	1111	1010
774	8	Construct a Johnson counter with Ten timing signals.		4				
775	8	Implement 4-bit ring counter using D FF.		5				
776	8	Design 4 bit twisted ring counter using JK Flip flop.		5				
777	8	For 4 bit Johson counter, If there are four flip flop namely FF0, FF1, FF2 and FF3 the which of the following is true?	A	1	output of FF3 is connected to the input of FF0	output of FF2 is connected to the input of FF1	output of FF3 is connected to the input of FF1	output of FF3 is connected to the output of FF0
778	8	For 4 bit Ring counter, If there are four flip flop namely FF1, FF2, FF3 and FF4 the which of the following is true?	B	1	output of FF3 is connected to the input of FF1	output of FF4 is connected to the input of FF1	output of FF4 is connected to the input of FF2	output of FF4 is connected to the output of FF1
779	8	Design synchronous counter with the following binary sequence: 0, 4,2,1,6,7 and repeat. Use JK flip-flops.		4				
780	8	A 4-bit serial in parallel out shift register is initially set to 1111. The data 1010 is applied to the input. After 3 clock cycles the output will be:		1				
781	8	On the fifth clock pulse, a 4-bit Johnson sequence is Q0 = 0, Q1 = 1, Q2 = 1, and Q3 = 1. On the sixth clock pulse, the sequence is _____.		1				
782	8	What is the time delay (td in μs) of an 8-bit serial-in/serial out shift register with a clock frequency of 4 MHz?		1				
783	8	Consider the following statements: 1. An n-bit ring counter is a mod-2n counter whereas an n-bit twisted ring counter is a mod-n counter. 2. The maximum modulus of ripple counter with seven flip-flops is 14. 3. Series counters have high speed but have more complex circuitry than parallel counters. 4. An n-bit counter has n number of flip-flops, 2 ⁿ number of states and also known as divide-by-2n counter. Which of the following statement/s is/are incorrect ?	D	1	3 and 4	2, 3 and 4	1, 2, and 4	1, 2, 3 and 4
784	8	Design a synchronous decade counter using JK-flip-flop.		5				
785	8	Design synchronous counter with the following binary sequence: 0, 4,2,1,6,7 and repeat. Use positive edge triggered T flip-flops.		4				
786	8	Design a synchronous counter which goes through sequence 0, 2, 4, 6, 7, 5, 3, 1, 0, ... using positive edge triggered T flip-flops		3				

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
787	8	Draw the state diagram and state table of BCD ripple counter, develop its logic diagram, and explain its operation using reset input. (No need of waveforms) Use negative edge triggered JK flip-flops.		4				
788	8	Draw 4-Bit Asynchronous Ripple Up/Down Counter using Negative Edge triggered T Flip-Flop		2				
789	8	The shift register shown in figure is initially loaded with the bit pattern 0011. Subsequently the shift register is clocked with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). Clock pulses taken by the shift register to get the content 0011 again are _____ 	C	1	3rd	4th	5th	6th
790	8	In a Johnson's counter, all the negative triggered J-K flip flop are used. Initially all the flip flops are in reset condition and the outputs are Q3 Q2 Q1 Q0 = 0000. What are the outputs of flip flops after the fifth negative going pulse ?	D	1	0101	1000	0010	1110
791	8	Consider the partial implementation of a 2 bit counter using T-flip flop following the sequence 0-2-3-1-0, as shown below : To complete the circuit the input X should be 	C	1	Q'2	Q2 + Q1	Q1 XOR Q2	Q1 XNOR Q2
792	8	Design decade counter using JK-FF which is clocked such that each flip-flop in the counter is triggered at the same time.		3				
793	8	How many flip flops are required to design mod-20 counter?	A	1	5	6	10	20
794	8	Design a shift register which gives double number of states from the number of flipflops connected with its state diagram.		4				
795	8	Design a mod-11 asynchronous counter using JK FF with necessary waveforms.		5				
796	9	ROM is made up of _____	DECODER & OR gate	1	NAND & OR gate	NOR & DECODER	DECODER & OR gate	NAND & DECODER
797	9	which of following is non-volatile memory ?	EEPROM	1	RAM	DRAM	EEPROM	SRAM
798	9	ROM has the capability to perform _____	read operation only	1	write operation only	read operation only	both write & read operation	erase operation
799	9	The ROM , which has to be custom built by factory, known as _____	mask ROM	1	EEPROM	MASK ROM	EPROM	PROM
800	9	How does PLA differ from a ROM		2				
801	9	Illustrate the mask rom and compare with it to EPROM		2				
802	9	Illustrate architecture of FPGA & its Application in modern times .		2				
803	9	EPROM contents can be erased by exposing it to infrared rays.	FALSE	1				
804	9	Elucidate that Full subtractor can be implemented using PAL.		2				
805	9	Is PROM differ from PLA & PAL ? if so, illustrate similarities & disimilarities among them.		2				
806	9	Illustrate PLA & compare it with PAL		2				
807	9	PLA can be used ...	to realise a combinational logic	1	as a microprocessor	as a dynamic memory	to realise a sequential logic	to realise a combinational logic
808	9	The minimum number of AND and OR Gate required for the implementation of Boolean function, $F1(A, B, C) = \sum m(0, 1, 3, 6, 7)$ $F2(A, B, C) = \sum m(3, 5, 6, 7)$ using PLA is:	C	1	3 AND and 2 OR	3 AND and 1 OR	4 AND and 2 OR	2 AND and 2 OR
809	9	PALs tend to execute _____ logic	SOP	1	SAP	SOP	PLA	SPD
810	9	Differentiate PAL with PLA.	The PLA has a programmable OR array and a programmable AND array, while the PAL only has a programmable AND Array	1	The PLA has a programmable OR array and a programmable AND array, while the PAL only has a programmable AND Array	The PAL has a programmable OR array and a programmable AND array, while the PLA only has a programmable AND array	The PAL has more possible product terms than the PLA.	PALs and PLAs are the same thing.
811	9	Product terms are the outputs of which type of gate within a PLD array?	AND	1	OR	XOR	AND	FLIP FLOP
812	9	Illustrate PLD 's with FPGA & state application of FPGA		2				
813	9	Implement the following using PLA $F1 = \sum(2, 4, 5, 10, 12, 13, 14)$ and $F2 = \sum(2, 9, 10, 11, 13, 14, 15)$.		4				
814	9	Implement the following functions using 8×3 PROM. $F1(m, n, o) = m'no' + m'n + o$ $F2(m, n, o) = m'no + mn'o + m'o$ $F3(m, n, o) = n'o + mo$		3				
815	9	illustrate neat diagram of an architecture of Field programming gate array & state their advantage over other Programmable logic devices		4				
816	9	A combinational circuit is defined by functions. $F1(A,B,C) = \sum(3, 5, 6, 7)$ & $F2(A,B,C) = \sum(0, 2, 4, 7)$ Implement the circuit with PLA having three inputs, four product term and two outputs		4				
817	9	Implement following functions using ROM. (Use suitable combinational circuit) $F1 = \sum m(1, 2, 3, 7)$ $F2 = \sum m(0, 2, 4, 6)$		3				
818	9	Using 8x4 ROM, realize the expressions $F1 = AB'C + ABC' + A'BC$, $F2 = A'B'C + A'BC' + AB'C'$, $F3 = A'B'C' + ABC$. Show the contents of all locations.		4				

Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
819	9	A combinational circuit is defined by the function $F1(A, B, C) = \sum m(0, 1, 2, 4)$ and $F2(A, B, C) = \sum m(0, 5, 6, 7)$. Implement the circuit with a PLA having 3 inputs, 4 product term & 2 outputs.		4				
820	9	A combinational circuit is defined by the function $F1(A, B, C) = \sum m(4, 5, 7)$ $F2(A, B, C) = \sum m(3, 5, 7)$. Implement the circuit with a PLA having 3 inputs, 3 product term & 2 outputs		4				
821	9	Implement the following equation by using PLA. $F1 = AB' + AC$ and $F2 = AC + BC$		4				
822	9	Tabulate the truth table for 8x4 ROM and implement the four boolean function listed below. Also minimize $A(X, Y, Z) = \sum(1, 3, 5, 6)$, $B(X, Y, Z) = \sum(0, 1, 6, 7)$, $C(X, Y, Z) = \sum(3, 5)$, $D(X, Y, Z) = \sum(1, 2, 4, 5, 7)$.		4				
823	9	Illustrate the construction in which a 3 bit addressable ROM, the following function are required. $h0 = \sum(0, 2, 5, 6)$ $h1 = \sum(0, 2, 4, 6, 7)$ $h2 = \sum(0, 2, 7)$ $h3 = \sum(1, 2, 3, 5, 7)$		4				
824	9	Illustrate the following function using PROM $f1 = \sum m(0, 1, 3, 4, 7)$ & $f2 = \sum m(1, 2, 5, 6)$		4				
825	9	Show the connection in PAL to yield the following $Y0 = A' + BC'$, $Y1 = A'B'CD + A'BC$, $Y2 = A'CD + AB$, $Y3 = A'B + C'D$		4				
826	9	The minimum number of AND and OR gates required for implementation of function using PLA is: $F1 = \sum m(0, 1, 3, 6, 7)$ $F2 = \sum m(3, 5, 6, 7)$		2				
827	9	Which of the following statement is correct about differentiating PAL and PLA? 1. PALs and PLAs are the same thing 2. The PLA has a programmable OR plane and a programmable AND plane, while the PAL only has a programmable AND plane 3. The PAL has a programmable OR plane and a programmable AND plane, while the PLA only has a programmable AND plane 4. The PAL has more possible product terms than the PLA.	A	1	only 2	only 3	only 4	only 1
828	9	Consider the following statement regarding PROM & EPROM 1) erasable programmable ROM using UV erasing is known as EPROM. 2) ROM that makes use of electrical voltage for erasing is known as electrically altered ROM 3) PROM can be programmed many times after fabrication. which of the above statement/s is / are correct ?	B	1	only 3	1 and 2	2 and 3	1 and 3
829	9	Implement the following boolean functions using suitable PAL: $w(A, B, C, D) = \sum m(1, 3, 4, 6, 9, 11, 12, 14)$ $x(A, B, C, D) = \sum m(1, 3, 4, 6, 9, 11, 12, 14, 15)$ $y(A, B, C, D) = \sum m(0, 2, 4, 6, 8, 12)$ $z(A, B, C, D) = \sum m(2, 3, 8, 9, 12, 13)$		4				
830	9	A combinational circuit is defined by functions. $F1(A, B, C) = \sum(0, 1, 2, 3, 6)$ and $F2(A, B, C) = \sum(3, 5, 7)$. Implement the circuit with PLA having three inputs, three product term and two outputs		5				
831	9	Implement FULL ADDER using Programmable Array Logic		3				
832	9	A combinational circuit is defined by functions: $X(A, B, C) = \sum m(2, 3, 5, 7)$ $Y(A, B, C) = \sum m(0, 1, 5)$ $Z(A, B, C) = \sum m(0, 2, 3, 5)$ Implement the circuit using PAL.		3				
833	9	FPGA stands for _____	C	1	Full Programmable Gate Array	Full Programmable Genuine Array	Field Programmable Gate Array	Field Programmable Gate Area
834	9	Determine the output of logic array given in the following figure. The X's represent connected link. 	B	1	$AB' + A'B$	0	1	$A'B' + AB$
835	9	Implement the following functions using PROM: $F1 = A(BC)' + ABC$ & $F2 = ABC + (AB)' + AC'$.		3				
836	9	Implement the following Boolean functions using PLA with 3 inputs, 3 product terms and 2 outputs. $F1 = \sum m(1, 3, 5)$ and $F2 = \sum m(5, 6, 7)$		4				
837	10	When used with an IC, what does the term “QUAD” indicate?	B	1	2 circuits	4 circuits	6 circuits	8 circuits
838	10	Give comparison of TTL and CMOS family		3				
839	10	Compare following for CMOS and TTL : Figure of merit, Noise margin, and Power dissipation		3				
840	10	Compare for CMOS and TTL: 1) Fan in 2) Noise Margin 3) Propagation Delay		3				
841	10	Define following terms: 1) Fan-out 2) Propagation delay 3) Power Dissipation		3				
842	10	_____ is the measure of maximum number of inputs that a single gate output can drive or accept.	Fan-in	1				
843	10	_____ means the maximum number of inputs that can be fed by a single output.	Fan-out	1				
844	10	Propagation delay is defined as _____	A	1	the time taken for the output of a gate to change after the inputs have changed	the time taken for the input of a gate to change after the outputs have changed	the time taken for the input of a gate to change after the intermediates have changed	the time taken for the output of a gate to change after the intermediates have changed
845	10	CMOS refers to _____	B	1	Continuous Metal Oxide Semiconductor	Complementary Metal Oxide Semiconductor	Centred Metal Oxide Semiconductor	Concrete Metal Oxide Semiconductor
846	10	Fan-in and Fan-out are the characteristics of _____	B	1	Registers	Logic families	Sequential Circuits	Combinational Circuits

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Digital Electronics (DE) Practice Book (Sem-III 2026)								
Note: This Practice Book is only for reference purpose. LJU Test question paper may not be completely set from Practice Book.								
Sr. No.	unit_number	question_text	answer_text	marks	option A	option B	option C	option D
847	10	Which logic family consumes the less power?	A	1	CMOS	TTL	PMOS	NMOS
848	10	Implement NAND gate using CMOS logic		2				
849	10	Implement NOR gate using CMOS logic		2				
850	10	TTL stands for _____	B	1	Totempole Logic	Transistor transistor logic	Transistor totempole Logic	None of Above
851	10	Which of the IC families have the largest fan-out?	A	1	CMOS	NMOS	PMOS	TTL
852	10	The figure of merit of a logic family is given by the product of	B	1	gain and bandwidth	propogation delay time and power dissipation	fan-out and propagation delay time	noise margin and power dissipation
853	10	Which equation is correct?	D	1	$V_{NL} = V_{IL(max)} + V_{OL(max)}$	$V_{NH} = V_{OH(min)} + V_{IH(min)}$	$V_{NL} = V_{OH(min)} - V_{IH(min)}$	$V_{NH} = V_{OH(min)} - V_{IH(min)}$
854	10	If a logic circuit has fan out 4 it means .	C	1	It has 4 inputs	It has 4 outputs	It can drive maximum 4 inputs	It can give 4 times the input
855	10	Classify the logic families. Give comparison with the advantages and disadvantages of CMOS and TTL logic families.		4				
856	10	The digital logic family which has minimum power dissipation is_____ .	A	1	CMOS	TTL	PMOS	NMOS
857	10	Compare CMOS and TTL in terms of Power dissipation, Fan-out, Noise margin and Propagation delay. Also implement NAND gate using TTL and CMOS logic.		5				
858	10	Implement NAND gate using logic family which has very low power dissipation per gate compare to other logic families.		2				
859	10	Design NOR gate using CMOS logic with its working principle and compare TTL and CMOS logic families, highlighting their unique features and characteristics.		3				
860	10	Implement 2 input NAND Gate using TTL Logic		2				
861	10	Compare CMOS over TTL logic families & illustrate NAND & NOT logic gates using CMOS.		3				
862	10	Define following terms: 1) Fan Out 2) Noise margin		2				